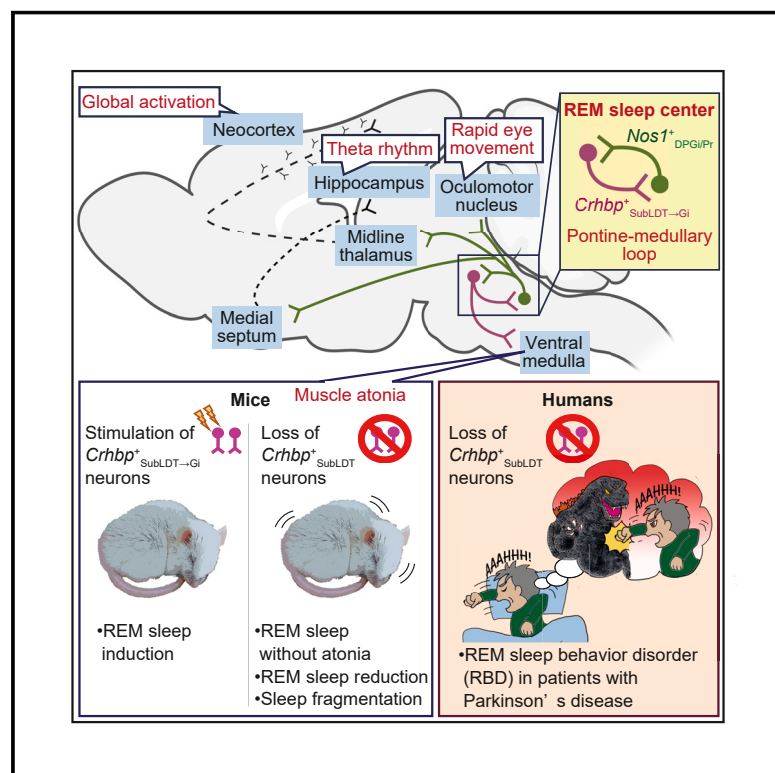


A pontine-medullary loop crucial for REM sleep and its deficit in Parkinson's disease

Graphical abstract



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In brief

A neuronal circuit originating in the brainstem pons and medulla strongly induces REM sleep, even in awake mice. Moreover, a subpopulation of neurons within this circuit is lost in patients with Parkinson's disease, a neurodegenerative disease characterized by severe REM sleep abnormalities.

Highlights

- $Crhbp^{+}_{SubLDT \rightarrow Gi}$ neurons promote REM sleep and muscle atonia
- $Nos1^{+}_{DPGi,Pr}$ neurons promote REM sleep and project to areas regulating forebrain activity
- The above 2 neuronal populations constitute a pontine-medullary loop circuit
- $CRHBP^{+}_{SubLDT}$ neurons are lost in human patients with Parkinson's disease and RBD



Article

A pontine-medullary loop crucial for REM sleep and its deficit in Parkinson's disease

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SUMMARY

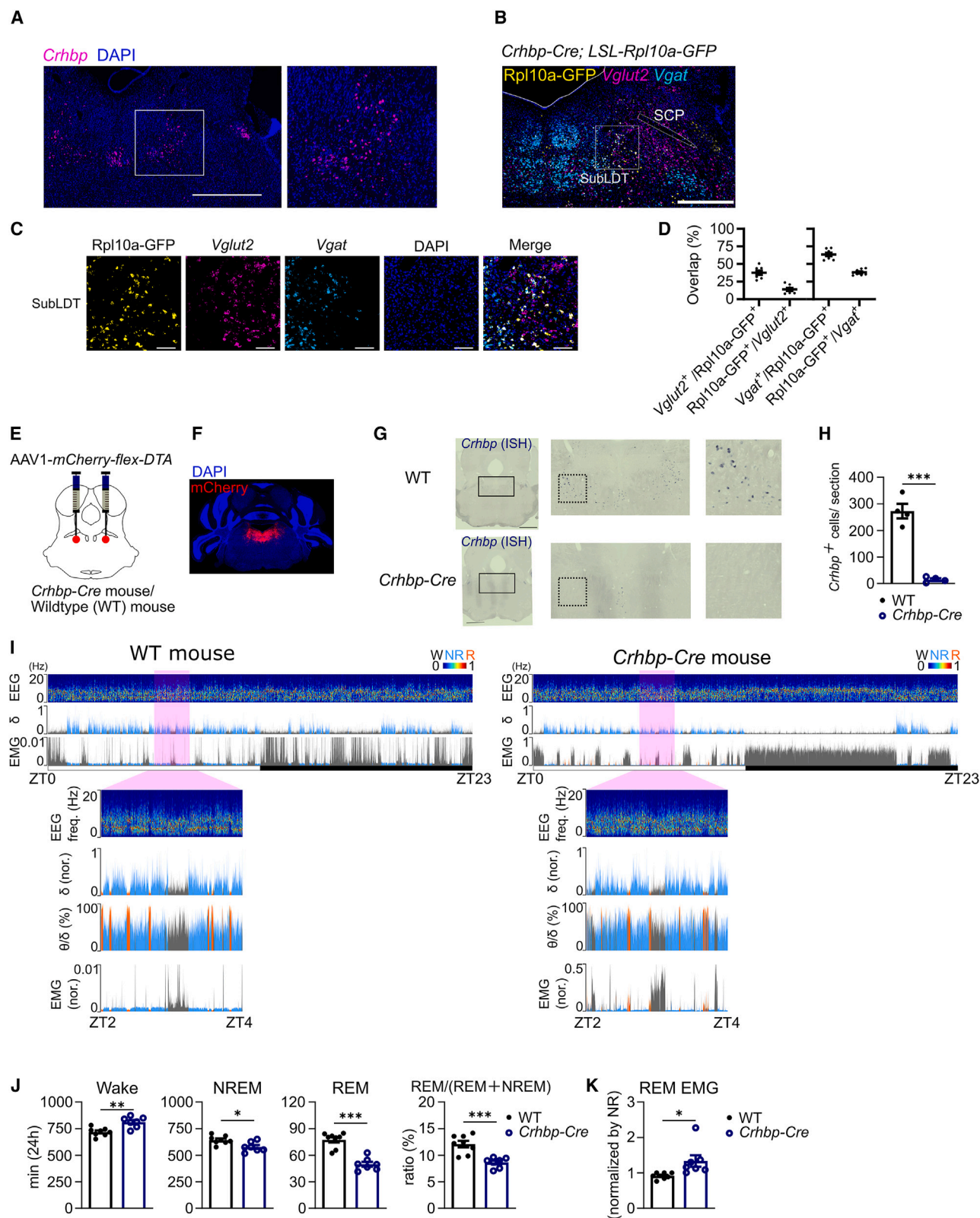
Identifying the properties of the rapid eye movement (REM) sleep circuitry and its relation to diseases has been challenging due to the neuronal heterogeneity of the brainstem. Here, we show in mice that neurons in the pontine sublaterodorsal tegmentum (SubLDT) that express corticotropin-releasing hormone-binding protein (*Crhbp*⁺ neurons) and project to the medulla promote REM sleep. Within the medullary area receiving projections from *Crhbp*⁺ neurons, neurons expressing nitric oxide synthase 1 (*Nos1*⁺ neurons) project to the SubLDT and promote REM sleep, suggesting a positively interacting loop between the pons and the medulla operating as a core REM sleep circuit. *Nos1*⁺ neurons also project to areas that control wide forebrain activity. Ablating *Crhbp*⁺ neurons reduces sleep and impairs REM sleep atonia. In Parkinson's disease patients with REM sleep behavior disorders, CRHBP-immunoreactive neurons are largely reduced and contain pathologic α -synuclein, providing insight into the mechanisms underlying the sleep deficits characterizing this disease.

INTRODUCTION

Mammalian sleep comprises rapid eye movement (REM) sleep (REMS) (or paradoxical sleep) and non-REM (NREM) sleep (NREMS). NREMS is characterized by high-amplitude slow-wave oscillations in the electroencephalogram (EEG) that reflect synchronous transitions of cortical neurons between depolarized up states and hyperpolarized down states. By contrast, REMS is characterized by increased average firing rates of cortical neurons, fast-frequency oscillations and theta waves in the EEG, and loss of skeletal muscle activity (muscle atonia). The pontine tegmentum in the brainstem is a key structure for REMS regulation.¹ Studies in cats implicated the peri-locus coeruleus (LC) α as a region within the pontine tegmentum crucial

for inducing REMS.^{2–5} Subsequent analyses suggested that the sublaterodorsal nucleus (SLD) in rodents is homologous to the cat peri-LC- α .^{6–8} Within the SLD, inhibition of glutamatergic neurotransmission reduces REMS amount and impairs muscle atonia.^{9–11} The glutamatergic neurons in this region are highly heterogeneous, however, and neurons that promote NREMS or wakefulness are intermingled.^{12–14} Whereas the entity of the neurons in SLD responsible for REMS remains obscure, recent studies revealed the roles of neurons in other areas in promoting REMS, namely the melanin-concentrating hormone-producing neurons in the lateral hypothalamus,^{15–17} the GABAergic neurons in the ventral medulla,¹⁸ and dopaminergic neurons in the ventral tegmental area.¹⁹ While these findings imply that REMS is under control of a disperse network of neurons distributed





(legend on next page)

throughout the brain rather than by a particular central circuit, notably, ablation/chronic inactivation of these neurons has limited effects on REMS amount.^{11,16,20,21} By contrast, lesion of the pontine area leads to an almost complete loss of REMS,^{3,5} suggesting that SLD indeed plays a central role. Thus, it is crucial to identify the neuronal subtype in the SLD that evokes REMS and reveal its circuitry.

Recently, deficits in REMS, particularly REMS behavior disorder (RBD), have drawn much attention due to their associations with synucleinopathies. RBD is a parasomnia characterized by dream-enacting body movements due to dysregulation of muscle atonia during REMS.^{22,23} Cohort studies indicate that RBD is a prodromal marker of synucleinopathies, including Parkinson's disease (PD).^{24–26} In addition to RBD, PD patients often suffer from excessive sleepiness and insomnia,²³ and sleep deficits in PD patients are characterized by progressive reductions in both REMS and NREMS.²⁷ Considering that advanced stages of synucleinopathies frequently accompany cognitive impairment and that REMS is required for memory consolidation,^{28–30} it is crucial to elucidate the cause of the sleep deficits in synucleinopathies. While neuronal degeneration and Lewy bodies, aggregates of α -synuclein, are observed in the pontine tegmentum of PD patients,^{26,31} the lack of molecular markers for neurons that control sleep has hampered research addressing their precise role in sleep-related symptoms.

In this study, based on the transcriptomic analysis of the pontine tegmentum,¹³ we identified a particular subtype of neurons that regulates REMS/NREMS and muscle atonia during REMS. Establishment of the genetic access to the REMS-promoting neurons in SLD enabled us to further examine their distribution, upstream and downstream circuits, and relevance to sleep disorders in humans.

RESULTS

Generation of *Crhbp*-Cre mice

Within the SLD, glutamatergic neurons that are maximally active during REMS are mainly detected in the sublateralodorsal tegmentum (SubLDT), or the rostral part of SLD that contains cholinergic neurons.^{32,33} Here, based on our previous transcriptomic analyses,¹³ we focused on corticotropin-releasing hormone-binding protein (*Crhbp*). *In situ* hybridization revealed that *Crhbp* is expressed in the SubLDT and, to a lesser extent, the laterodorsal tegmental nucleus (LDT) located dorsally (Figure 1A). *Crhbp* was also expressed in the ventral tegmental nu-

cleus (VT) located medially (Figure 1A). We generated a *Crhbp*-Cre (*Crhbp*-P2A-Cre-pA) knockin mouse (Figures S1A and S1B). When we crossed *Crhbp*-Cre mice with a reporter mouse that Cre-dependently expresses green fluorescent protein (GFP) fused to Rpl10a (*LSL-Rpl10a-GFP*), GFP signals were detected in the SubLDT (Figure 1B) and largely recapitulated the endogenous *Crhbp* expression pattern (Figures S1C–S1E). *Crhbp*-expressing neurons in the SubLDT (*Crhbp*⁺_{SubLDT} neurons) either co-expressed vesicular GABA transporter (*Vgat*: *Slc32a1*) or vesicular glutamate transporter 2 (*Vglut2*: *Slc17a6*) (Figures 1B–1D), indicating that these neurons are either GABAergic or glutamatergic.

Ablation of *Crhbp*⁺_{S,L,V} neurons reduces sleep and impairs muscle atonia during REMS

To test the involvement of *Crhbp*⁺ neurons in sleep regulation, we ablated the *Crhbp*⁺ neurons in the SubLDT and adjacent areas, the LDT, and VT (*Crhbp*⁺_{S,L,V} neurons) (Figures 1E–1K and S1F–S1P). We injected an adeno-associated virus (AAV) carrying Cre-dependent diphtheria toxin fragment-A (DTA) (AAV.1-*Ef1a-mCherry-flex-DTA*) into this area in *Crhbp*-Cre or wild-type (WT) control mice. mCherry expression confirmed the area of AAV infection (Figure 1F). Compared with the WT control group, the number of *Crhbp*⁺_{S,L,V} neurons, assessed by *in situ* hybridization for *Crhbp* mRNA, was reduced in the *Crhbp*-Cre group (Figures 1G and 1H), confirming that DTA-mediated ablation occurred Cre-dependently. We recorded the EEG and the electromyogram (EMG) to measure sleep. Ablation of *Crhbp*⁺_{S,L,V} neurons reduced the time spent in both REMS and NREMS with a concomitant increase in wakefulness (Figures 1I, 1J, and S1F–S1P). In addition, the *Crhbp*-Cre group exhibited higher EMG integral values during REMS (Figures 1I–1K and S1M), indicating that these neurons also regulate muscle atonia during REMS. *Crhbp*-Cre; *Vglut2*^{flox/flox} mice, in which glutamatergic transmission is impaired in *Crhbp*⁺ neurons, similarly exhibited reduced REMS and NREMS and the development of REMS without atonia (Figures S1Q–S1U). Thus, the glutamatergic *Crhbp*⁺_{S,L,V} neurons are involved in positively regulating both REMS and NREMS and the expression of muscle atonia during REMS.

We next estimated whether *Crhbp*⁺_{SubLDT} neurons are active during REMS by combining immunohistochemistry for c-fos, a marker for neural excitation, with the flowerpot method (Figures S1V–S1X). The flowerpot method largely reduces the amount of REMS, leading to a large rebound of

Figure 1. Ablation of *Crhbp*⁺_{S,L,V} neurons reduces REMS and NREMS and impairs muscle atonia during REMS

(A) Distribution of *Crhbp* mRNA in SubLDT (scale bar: 1 mm; magenta: *Crhbp* [*in situ* hybridization, ISH], blue: 4', 6-diamidino-2-phenylindole, DAPI). (B–D) Histologic verification of the neurotransmitters of *Crhbp*⁺_{SubLDT} cells using *Crhbp*-Cre; *LSL-Rpl10a-GFP* mice. (Scale bars: 0.5 mm in B and 0.1 mm in C; yellow: Rpl10a-GFP, magenta: *Vglut2* [ISH], cyan: *Vgat* [ISH], blue: DAPI). Data represent mean $\pm$ SEM. *n* = 8 sections/group, obtained from 4 mice. (E and F) AAV-encoding mCherry and Cre-dependent DTA were microinjected into SubLDT, LDT, and VT of WT or *Crhbp*-Cre mice. mCherry was expressed in Cre-negative cells and served as an infection marker. (Red: mCherry, blue: DAPI). (G and H) Comparison of the number of *Crhbp*⁺_{S,L,V} neurons between AAV-infected WT and *Crhbp*-Cre mice by ISH for *Crhbp*. Purple depositions were developed by BCIP/NBT. Numbers of *Crhbp*⁺ cells in the solid squares were counted. Data represent mean $\pm$ SEM, *n* = 4 sections obtained from 4 mice. ****p* < 0.001 (unpaired *t* test). (I) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral in *Crhbp*-Cre or WT mouse, respectively. (J and K) Comparison of the amount of each vigilance state (J) or EMG during REMS (K) across 24 h between AAV-infected WT and *Crhbp*-Cre mice. Data represent mean $\pm$ SEM. *n* = 8 mice in WT group, *n* = 7 mice in *Crhbp*-Cre group. **p* < 0.05; ****p* < 0.001 (unpaired *t* test). See also Figures S1 and S7.

REMS.^{10,21,34,35} *Crhbp-Cre; LSL-Rpl10a-GFP* mice were divided into basal group (mice housed in their home cage), REMS-deprived group (mice subjected to the flowerpot method), or REMS-rebound group (mice subjected to the flowerpot method and subsequently returned to their home cage). As expected, the REMS-rebound group spent a greater amount of time in REMS compared with the other groups (Figure S1W). The ratio of c-fos- and Rpl10a-GFP-double-positive neurons to all Rpl10a-GFP-positive neurons in the SubLDT was higher in the REMS-rebound group compared with the other groups (Figure S1X), suggesting the activation of at least a subset of *Crhbp*⁺_{SubLDT} neurons during REMS.

Projection target of *Crhbp*⁺_{SubLDT} neurons

We examined the axonal projection patterns of *Crhbp*⁺_{SubLDT} neurons (Figure 2). We unilaterally injected an AAV that Cre-dependently expresses the presynaptic protein synaptophysin fused to GFP (AAV.1-*hSyn-flex-tdTomato-2A-synaptophysin-GFP*) into the SubLDT of *Crhbp-Cre* mice (Figures 2A and 2B). GFP signals were detected at the medial septum, basal forebrain (BF), mediodorsal thalamus, hypothalamus, dorsal deep mesencephalic nucleus (dDpMe), and wide areas of the medulla, including the gigantocellular reticular nucleus (Gi) and paragigantocellular nuclei (Figures 2C–2G).

Crhbp⁺_{SubLDT→Gi} neurons promote REMS

We tested whether *Crhbp*⁺_{SubLDT} neurons have distinct roles depending on their projections. We focused on Gi-projecting neurons because the ventral region of the Gi partially overlaps with the distribution of inhibitory neurons essential for muscle atonia during REMS.^{11,21} To manipulate *Crhbp*⁺_{SubLDT} neurons projecting to the Gi (*Crhbp*⁺_{SubLDT→Gi} neurons), we injected AAV-carrying flippase-dependent hM3Dq-YFP (AAV.8-*hSyn-fDIO-hM3Dq-YFP*) into the SubLDT and retrograde AAV-carrying Cre-dependent flippase (AAV.rg-*hSyn-DO-mCherry-DIO-flippase*), which efficiently transfects from axon terminals, into the Gi of *Crhbp-Cre* mice (Figures 3A and 3B). hM3Dq is a genetically engineered Gq-coupled receptor that induces neural excitation upon binding with its synthetic ligand clozapine-N-oxide (CNO). Most of the *Crhbp*⁺_{SubLDT→Gi} neurons were glutamatergic based on the co-expression of *Vglut2* (Figures S2A and S2B). When these neurons were chemogenetically activated during the light phase by CNO administration at zeitgeber time (ZT) 5, REMS was increased at the expense of wakefulness, resulting in an increased ratio of REMS to total sleep (Figures 3C, 3D, S2B, and S2C). REMS increase was due to both increased number of REMS episodes and their mean duration (Figures S2D and S2E), and the latency to enter the first REMS episode was largely reduced (Figure S2F), suggesting that these neurons contribute to both the initiation and maintenance of REMS. Chemogenetic activation of *Crhbp*⁺_{SubLDT→Gi} neurons also deepened muscle atonia during REMS (Figure S2G). Chemogenetic activation during the dark phase (ZT17) also resulted in increased REMS and a higher ratio of REMS to total sleep (Figures S2H–S2K). During the dark phase, chemogenetic activation also increased NREMS, which might be a second-

ary effect of REMS increase because basal sleep amount is low during the dark phase and chemogenetically induced REMS was preceded by NREMS. REMS was not increased when CNO was administered into *Crhbp-Cre* mice injected with AAV-carrying flippase-dependent jGCaMP7s (SubLDT: AAV.8-*hSyn-fDIO-jGCaMP7s*, Gi: AAV.rg-*hSyn-DO-mCherry-DIO-flippase*) (Figure S2L), suggesting that the observed effect was not due to non-specific effects of CNO or its derivative. Utilizing these mice expressing jGCaMP7s in the *Crhbp*⁺_{SubLDT→Gi} neurons (Figures 3E and 3F), we next measured the population activity of these neurons by measuring bulk Ca²⁺ transient across the spontaneous sleep-wake cycle by fiber photometry (Figures 3E–3I). The fiber was inserted above the SubLDT. Population activity was maximal during REMS (Figures 3G and 3H). At the transitions between each vigilance state, the fluorescent signals increased from NREMS to REMS and from NREMS to wake and decreased from wake to NREMS. The signals changed little during REMS to wake transitions (Figure 3I). Considering that activation of *Crhbp*⁺_{SubLDT→Gi} neurons did not increase wakefulness, the activation of these neurons during wakefulness was unexpected. Perhaps basal activity of *Crhbp*⁺_{SubLDT→Gi} neurons is important for limiting wakefulness because activation of these neurons reduced wakefulness. Alternatively, a small subpopulation of *Crhbp*⁺_{SubLDT→Gi} neurons might not be engaged in REMS regulation, and these neurons might be active during wakefulness. In the future, it is important to record activity of *Crhbp*⁺_{SubLDT→Gi} neurons at single-cell resolution.

Crhbp⁺_{SubLDT→BF} neurons promote NREMS

We examined the effect of manipulating *Crhbp*⁺_{SubLDT} neurons projecting to the BF (*Crhbp*⁺_{SubLDT→BF} neurons; Figures 4A–4D). Most of the *Crhbp*⁺_{SubLDT→BF} neurons were GABAergic based on the expression of *Vgat* (Figure 4J). When we chemogenetically activated these neurons (*Crhbp-Cre* mice, SubLDT: AAV.8-*hSyn-fDIO-hM3Dq-YFP*, BF: AAV.rg-*hSyn-DO-mCherry-DIO-flippase*), NREMS increased with a concomitant decrease in wakefulness and in the REMS to total sleep ratio in both the light phase (ZT5) and dark phase (ZT17) (Figures 4C, 4D, S2M–S2P, and S2R). In contrast to *Crhbp*⁺_{SubLDT→Gi} neurons, chemogenetic activation of *Crhbp*⁺_{SubLDT→BF} neurons did not affect muscle atonia during REMS (Figure S2Q). Population activity measured by fiber photometry (*Crhbp-Cre* mouse, SubLDT: AAV.8-*hSyn-fDIO-jGCaMP7s* + optic fiber, BF: AAV.rg-*hSyn-DO-mCherry-DIO-flippase*) (Figures 4E–4I) was highest during wakefulness (Figures 4G and 4H). At the transitions between each vigilance state, the fluorescent signal increased from NREMS to REMS, from NREMS to wake, and from REMS to wake, whereas it decreased from wake to NREMS (Figure 4I). Considering that chemogenetic activation of *Crhbp*⁺_{SubLDT→BF} neurons induces NREMS and that *Crhbp*⁺_{SubLDT→BF} neurons are active during wakefulness, these neurons might be involved in restraining wakefulness.

We examined whether *Crhbp*⁺_{SubLDT→Gi} and *Crhbp*⁺_{SubLDT→BF} neurons overlap with each other by labeling each subpopulation with distinct retrograde tracers (*Crhbp-Cre* mice, Gi: AAV.rg-*hSyn-DIO-GFP*, BF: AAV.rg-*CAG-flex-tdTomato*). There was

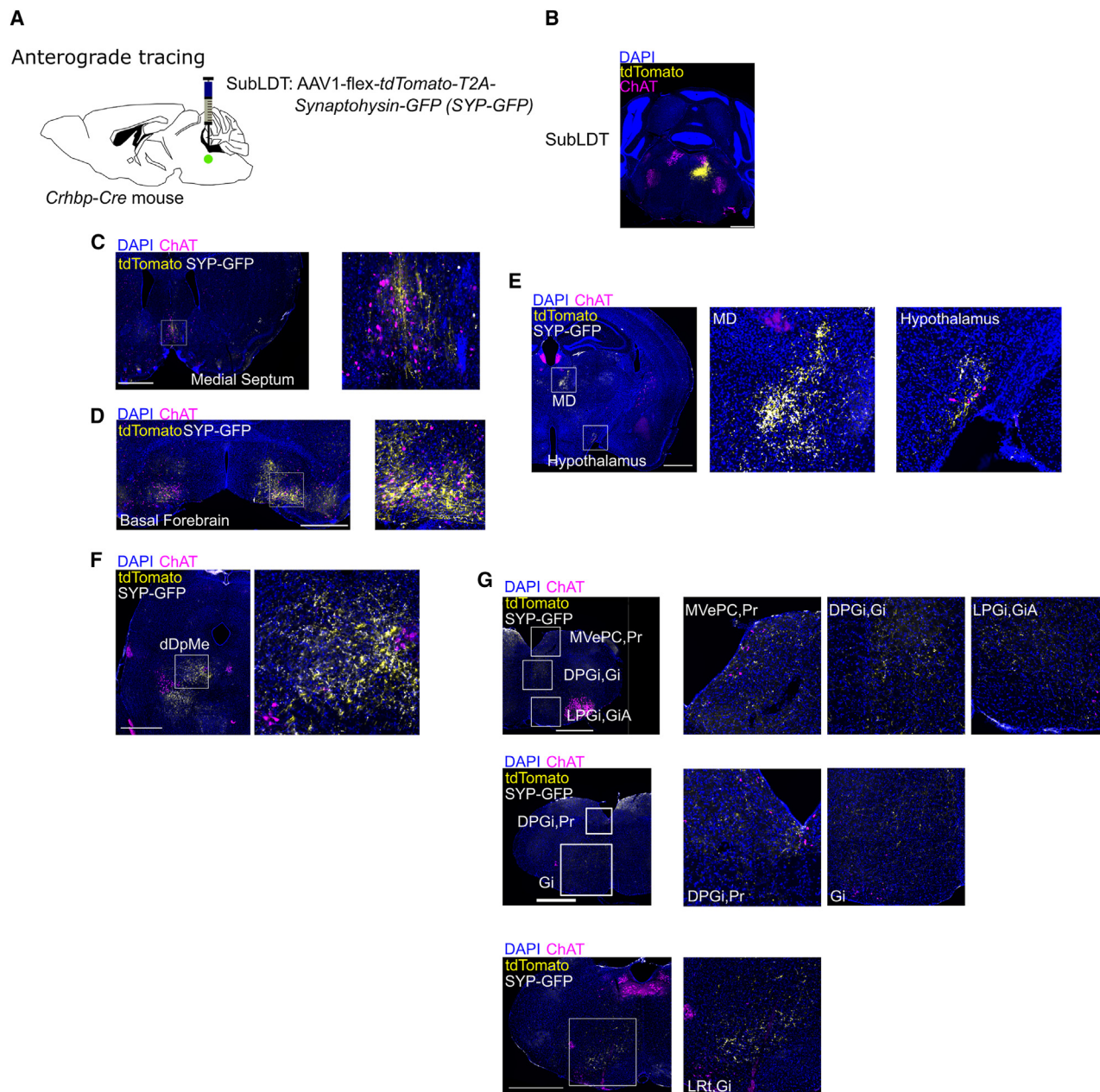


Figure 2. Projection targets of $Crhbp^{+}_{SubLDT}$ neurons

(A) AAV-carrying Cre-dependent tdTomato and synaptophysin-GFP were unilaterally microinjected into SubLDT of $Crhbp$ -Cre mice.

(B) AAV-derived signals were detected in SubLDT. (Scale bar: 1 mm; yellow, tdTomato; magenta, ChAT; blue, DAPI).

(C–G) Synaptophysin-GFP signals were detected in medial septum (C), basal forebrain (D), mediodorsal thalamus (MD) and hypothalamus (E), dorsal deep mesencephalic nucleus (dDpMe) (F), and wide areas of the medulla including gigantocellular reticular nucleus (Gi), medial vestibular nucleus, parvicellular part (MVePC), prepositus hypoglossal nucleus (Pr), dorsal paragigantocellular nucleus (DPGi), lateral paragigantocellular nucleus (LPGi), gigantocellular reticular nucleus, alpha part (GiA), and lateral reticular nucleus (LRt) (G).

no overlap between the two tracers (Figure S2S), suggesting that $Crhbp^{+}_{SubLDT \rightarrow Gi}$ and $Crhbp^{+}_{SubLDT \rightarrow BF}$ neurons are distinct populations.

Chemogenetic activation of $Crhbp^{+}_{SubLDT}$ neurons without any projection specificity ($Crhbp$ -Cre mouse, SubLDT: AAV.2-*hSyn*-

DIO-hM3Dq-mCherry) produced similar results (Figures 4K–4M), presumably due to the dominant effect of the $Crhbp^{+}_{SubLDT \rightarrow BF}$ neurons. Thus, $Crhbp^{+}_{SubLDT}$ neurons comprise at least 2 distinct populations of neurons that promote either REMS or NREMS, depending on their axonal projection patterns.

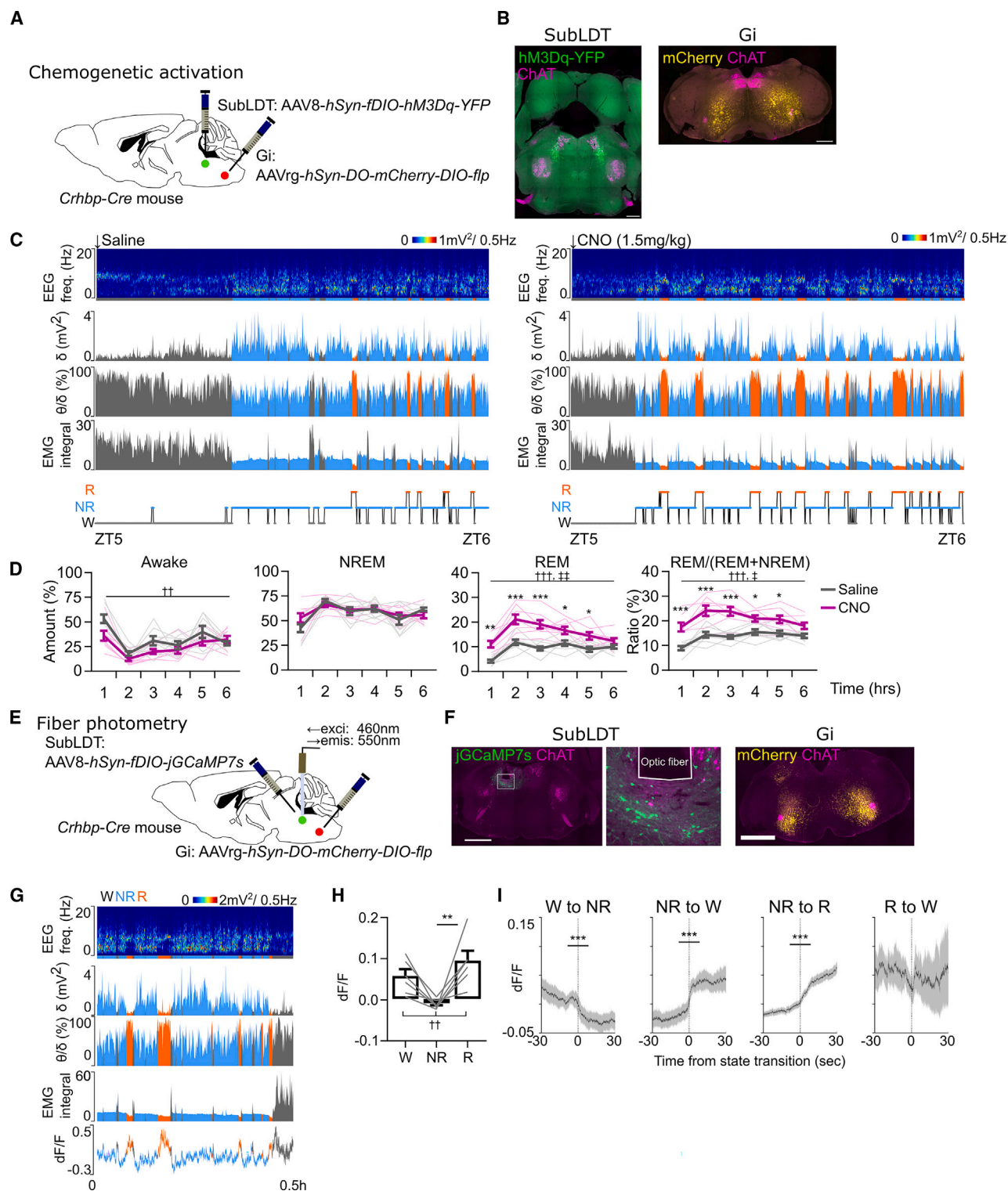


Figure 3. *Crhbp*⁺ SubLDT → Gi neurons promote REMS

(A) Retrograde AAV-encoding Cre-dependent flippase was microinjected into Gi, and AAV-encoding flippase-dependent hM3Dq-YFP was microinjected into SubLDT of *Crhbp-Cre* mice.

(B) hM3Dq-YFP and mCherry were detected in SubLDT and Gi, respectively. mCherry, expressed independently of Cre, served as an infection marker of AAV-carrying Cre-dependent flippase (scale bar: 0.5 mm; green, hM3Dq-YFP; magenta, ChAT; yellow, mCherry).

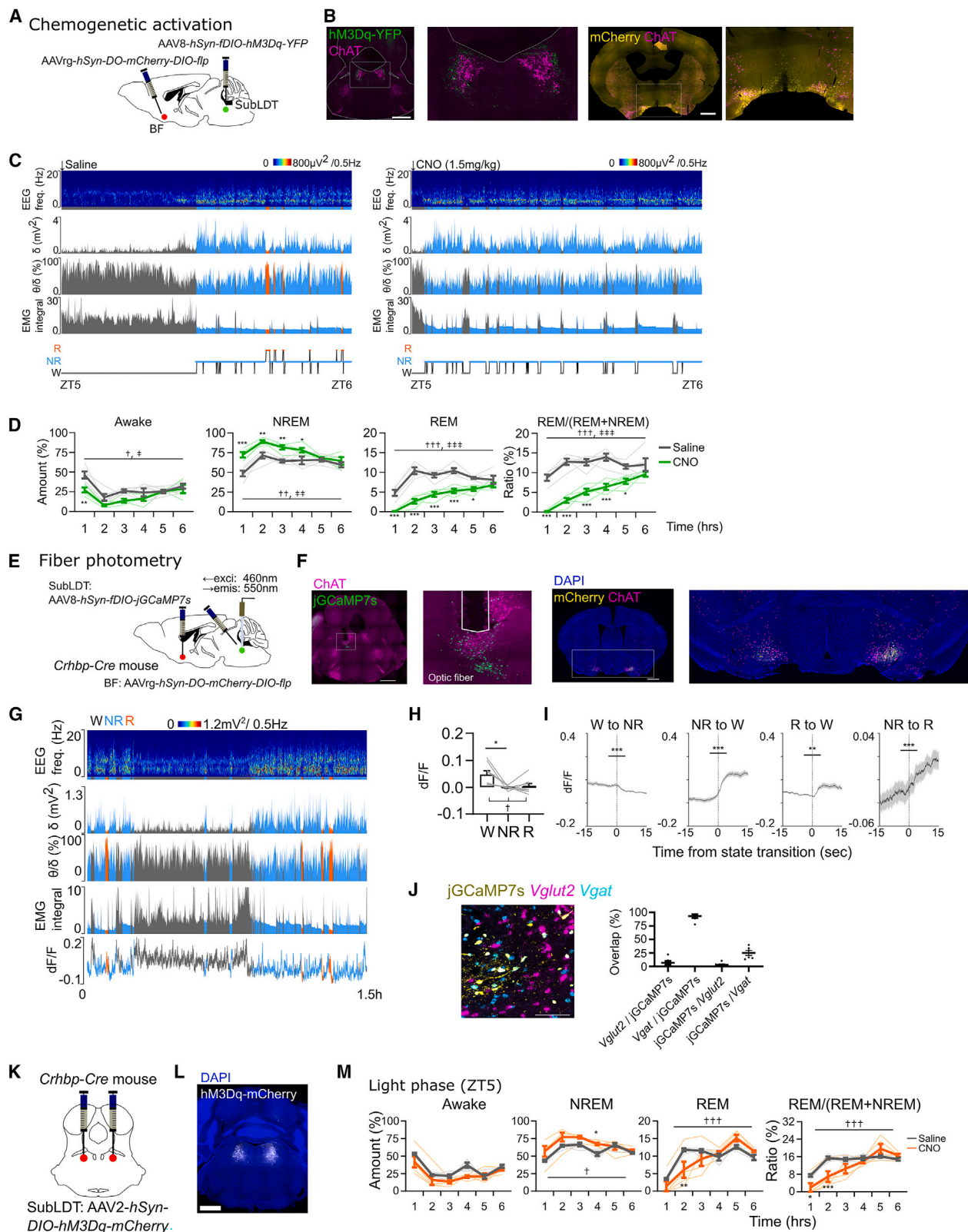
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***Camklla*⁺_{DPGi,Pr} neurons promote REMS via projection to SubLDT**

To explore the neurons upstream of the REMS-promoting *Crhbp*⁺_{SubLDT→Gi} neurons, we performed rabies virus (RV) tracing (Figure 5). A mixture of AAV-carrying Cre-dependent avian receptor TVA (AAV.10-*Ef1a-flex-TVA-mCherry*) and AAV-carrying Cre-dependent rabies glycoprotein (RG) (AAV.10-*CAG-flex-RG*), necessary for the retrograde trans-synaptic spread of RV, was injected into the SubLDT of *Crhbp-Cre* mice. Pseudotyped G-deleted RV carrying GFP (*SADΔG-GFP* [EnvA]) was injected into the Gi (Figures 5A and 5B). GFP-positive neurons were detected in the dDpMe, lateral paraventricular nucleus (LPGi), dorsal paraventricular nucleus (DPGi), and prepositus hypoglossal nucleus (Pr) (Figure 5C), areas involved in REMS regulation.^{12,13,18,34–39} GABAergic neurons in the dDpMe inhibit REMS,¹² which can be partly explained by our findings that dDpMe neurons project to *Crhbp*⁺_{SubLDT→Gi} neurons. In addition, putative glutamatergic neurons in the LPGi fire maximally during REMS,⁴⁰ and might thus contribute to activate *Crhbp*⁺_{SubLDT→Gi} neurons during REMS. Recent studies demonstrated that GABAergic neurons within the dorsomedial medulla, including those in the DPGi and the Pr, promote REMS, and another subset of glutamatergic neurons in the DPGi contribute to REM during REMS without affecting the amount of REMS.^{36,38} Based on these reports, direct projections from the DPGi/Pr to the REMS-promoting neurons in the SubLDT were unexpected. The GFP-labeled cells in the DPGi/Pr, i.e., the DPGi/Pr neurons projecting to *Crhbp*⁺_{SubLDT→Gi} neurons, were either glutamatergic or GABAergic, based on the expression of *Vglut2* and *Vgat* (Figure 5D). We also found that the DPGi/Pr neurons projecting to *Crhbp*⁺_{SubLDT→Gi} neurons can be labeled by AAV carrying the *Camklla* promoter (Figure 5D). Consistently, when we injected AAV.8-*Camklla-GFP* into the DPGi/Pr of WT mice, GFP-labeled axons were detected in the SubLDT (Figure 5E), indicating that DPGi/Pr neurons projecting to the SubLDT can be labeled with the *Camklla* promoter (hereafter, *Camklla*⁺_{DPGi,Pr} neurons). To functionally inhibit the input from *Camklla*⁺_{DPGi,Pr} neurons to SubLDT, we injected AAV-carrying blue light-activated chloride channel *SwiChR++-YFP* into the DPGi/Pr and placed an optic fiber above the right SubLDT while neurons of the left SubLDT were ablated by expressing DTA (WT mice, DPGi/Pr: AAV.5-*Camklla-SwiChR++-YFP*, left

SubLDT: AAV.1-*Ef1a-mCherry-flex-DTA*/AAV.1-*hSyn-GFP-Cre*, right SubLDT: optic fiber) (Figures 5F and 5G). During the light period, starting from ZT6, a 1-s pulse of blue light was delivered every 20 min for 4 h. In the control condition, no blue light was delivered. The time spent in REMS was reduced during the 4 h of blue light illumination due to the reduced mean episode duration (Figures 5H, 5I, S3A, and S3B). A different stimulation protocol (a 1-s pulse of blue light delivered every 50 s) also resulted in reduced REMS (Figures S3C and S3D). In addition, bilateral optogenetic inhibition (without unilateral ablation) with either stimulation protocol (1-s pulse every 20 min or 1-s pulse every 50 s) also resulted in reduced REMS (Figures S3E–S3I). We confirmed that neither unilateral ablation of SubLDT neurons plus insertion of the optic fiber to the contralateral side nor bilateral insertion of optic fibers above SubLDT affects basal sleep amount (Figure S3J). Conversely, chemogenetic activation of *Camklla*⁺_{DPGi,Pr} neurons projecting to SubLDT (WT mice, DPGi/Pr: AAV.8-*Camklla-DIO-hM3Dq-mCherry*, SubLDT: AAV.rg-*mSyn-BFP-Cre*) either in the light (ZT5) or dark (ZT17) phase increased REMS (Figures 5J–5M and S3K–S3M). Contrastingly, chemogenetic activation of *Camklla*⁺_{DPGi,Pr} neurons projecting to the superior colliculus (SC) (WT mice, SC: AAV.rg-*mSyn-BFP-Cre*, DPGi/Pr: AAV.8-*Camklla-DIO-hM3Dq-mCherry*) did not increase the time spent in REMS (Figures S3N–S3Q). These results suggest that *Camklla*⁺_{DPGi,Pr} neurons induce REMS, at least partly through projections to SubLDT. We also found that chemogenetic activation of *Camklla*⁺_{DPGi,Pr} neurons without any projection specificity (WT mice, DPGi/Pr: AAV.8-*Camklla-hM3Dq-mCherry*) strongly induced REMS in both the light (ZT5) and dark (ZT17) phase, whereas chemogenetic inhibition (WT mice, DPGi/Pr: AAV.8-*Camklla-hM4Di-mCherry*) suppressed REMS (Figures 5N–5P and S4A–S4L). The latency to enter REMS following chemogenetic activation of *Camklla*⁺_{DPGi,Pr} neurons was very short (Figure S4G), suggesting that these neurons have robust REMS-inducing effects. Fiber photometry recordings of *Camklla*⁺_{DPGi,Pr} neurons (WT mice, DPGi/Pr: AAV.9-*Camklla-GCaMP6s*) detected maximal Ca^{2+} signals during REMS (Figures 5Q–5T and S4M), and c-fos immunohistochemistry combined with the flowerpot method suggested that *Camklla*⁺_{DPGi,Pr} neurons are activated during rebound REMS (Figures S4N and S4O). Thus, at least a subpopulation of *Camklla*⁺_{DPGi,Pr} neurons is maximally active during

- (C) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and hypnogram after administration of saline or CNO.
- (D) Comparison of the effect of saline and CNO (1.5 mg/kg) administered at ZT5 on the amount of each vigilance state. Horizontal axes indicate time after drug administration. Data represent mean ± SEM. $n = 8$ mice/group. * $p < 0.05$; **,†† $p < 0.01$; ***,†††,†††† $p < 0.001$ (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).
- (E) Retrograde AAV-encoding Cre-dependent flippase was microinjected into Gi, and AAV-encoding flippase-dependent jGCaMP7s was microinjected into SubLDT of *Crhbp-Cre* mice. The optic fiber was inserted above SubLDT.
- (F) jGCaMP7s and mCherry were detected in SubLDT and Gi, respectively. mCherry, expressed independently of Cre, served as an infection marker of the AAV-carrying Cre-dependent flippase (scale bar: 1 mm; green, jGCaMP7s; magenta, ChAT; yellow, mCherry).
- (G) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and photometry signal.
- (H) Comparison of fluorescent signals among each vigilance state. Data represent mean ± SEM. $n = 6$ mice. **,†† $p < 0.01$, (one-way ANOVA followed by Tukey's test, † indicates significant difference among vigilance states in one-way ANOVA, and * indicates significance in Tukey's test).
- (I) Photometry signals aligned to vigilance state transitions. Data represent mean ± SEM. Wake to NREMS, $n = 20$ episodes obtained from 6 mice; NREMS to wake, $n = 18$ episodes obtained from 6 mice; NREMS to REMS, $n = 45$ episodes obtained from 6 mice; REMS to wake, $n = 6$ episodes obtained from 6 mice. *** $p < 0.001$ (paired Student's t test).
- See also Figures S2 and S7.



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REMS and is essential for REMS induction. *Camklla*⁺_{DPGi,Pr} neurons were either excitatory or inhibitory based on their expression of *Vglut2* or *Vgat* (Figures S4P and S4Q). We next tested whether both types of *Camklla*⁺_{DPGi,Pr} neurons are involved in the promotion of REMS. Chemogenetic inhibition of inhibitory *Camklla*⁺_{DPGi,Pr} neurons (*Vgat*-Cre; AAV.8-*Camklla*-DIO-hM4Di-mCherry) reduced REMS (Figures S4T and S4U), supporting that a subpopulation of *Camklla*⁺_{DPGi,Pr} neurons overlaps with GABAergic REMS-promoting neurons in the previous investigation.³⁸ In addition, chemogenetic inhibition of the excitatory *Camklla*⁺_{DPGi,Pr} neurons (*Vglut2*-Cre; AAV.8-*Camklla*-DIO-hM4Di-mCherry) also reduced REMS (Figures S4R and S4S). The inhibition of excitatory neurons also decreased NREMS. Thus, both excitatory and inhibitory *Camklla*⁺_{DPGi,Pr} neurons contribute to the strong REMS-promoting effect.

Nos1⁺_{DPGi,Pr} neurons strongly promote REMS and project to multiple areas involved in REMS

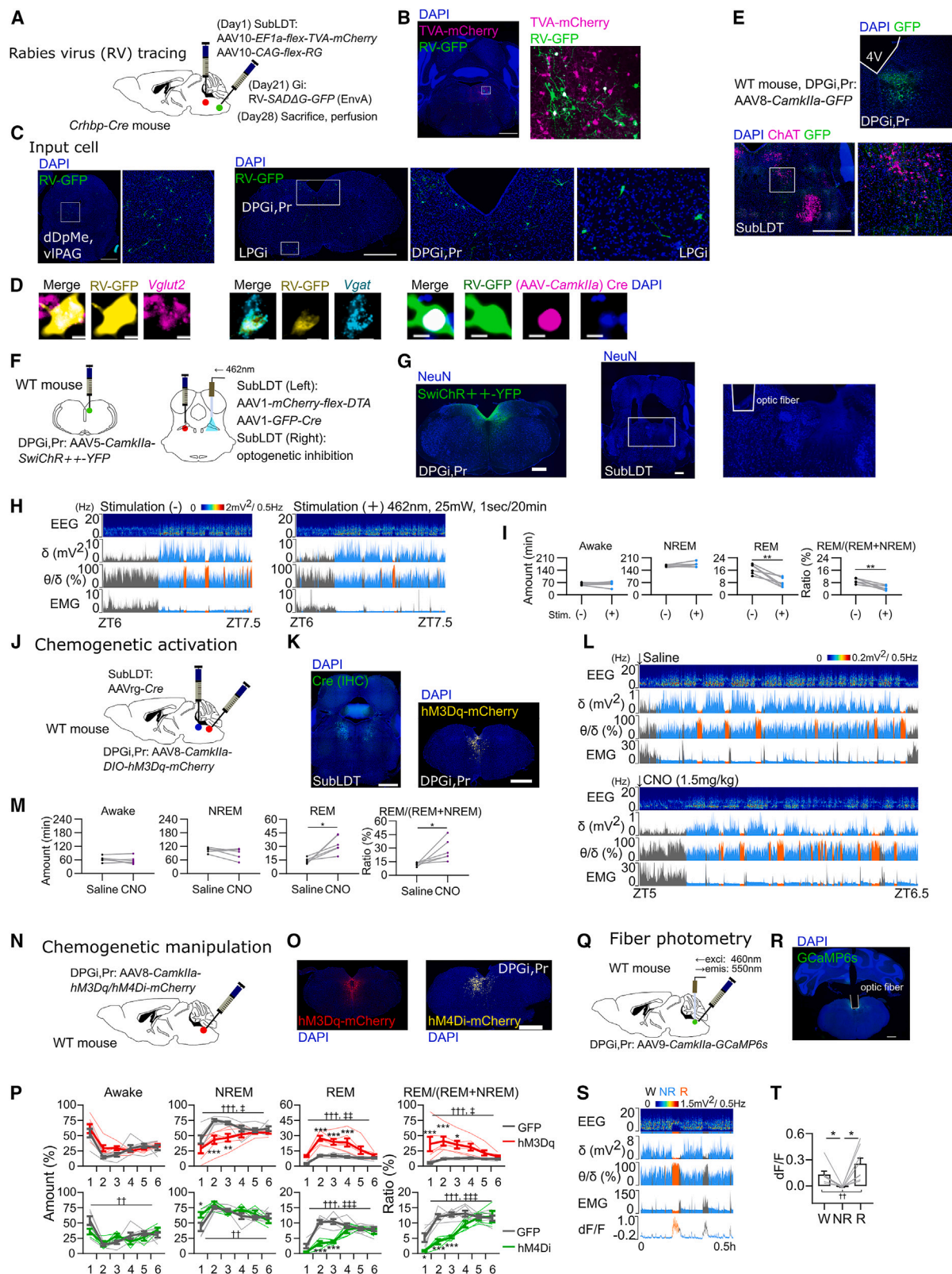
Within the DPGi/Pr, we next focused on nitric oxide synthase 1 (*Nos1*), as we noticed that its expression is enriched in the DPGi/Pr (<http://mouse.brain-map.org>). We examined whether *Camklla*⁺_{DPGi,Pr} neurons co-express *Nos1*. We injected a mixture of AAV-carrying Cre-dependent tdTomato (AAV.1-CAG-*flex*-tdTomato) and AAV-carrying GFP under control of the *Camklla*⁺ promoter (AAV.8-*Camklla*-GFP) into the DPGi/Pr of *Nos1*-Cre mice (Figure S5A). A large portion of the tdTomato-positive *Nos1*⁺_{DPGi,Pr} neurons overlapped with *Camklla*⁺_{DPGi,Pr} neurons (Figure S5A). *Nos1*⁺_{DPGi,Pr} neurons were either glutamatergic or GABAergic (Figure S5B) and projected to *Crhbp*⁺_{SubLDT→Gi} neurons based on RV tracing (Figure S5C).

To chemogenetically activate *Nos1*⁺_{DPGi,Pr} neurons, we injected AAV-carrying Cre-dependent hM3Dq-mCherry (AAV.2-*hSyn*-DIO-hM3Dq-mCherry) into the DPGi/Pr of *Nos1*-Cre mice (Figures 6A and 6B). When we activated these neurons at ZT5, REMS was increased at the expense of NREMS, resulting in an increased ratio of REMS to total sleep (Figures 6C and 6D). REMS increase was due to increased number of REMS episodes (Figures 6E–6G). For NREMS, episode number was increased, whereas their duration was decreased (Figures 6E–6G), suggesting an increase in transitions from NREMS to REMS. Chemogenetic inhibition of *Nos1*⁺_{DPGi,Pr} neurons (*Nos1*-Cre mice, DPGi/Pr: AAV.8-*hSyn*-DIO-hM4Di-mCherry) reduced the amount of REMS and extended the mean duration of NREMS (Figures S5D–S5I). Fiber photometry recordings of *Nos1*⁺_{DPGi,Pr} neurons (*Nos1*-Cre mice, DPGi/Pr: AAV.9-*hSyn*-*flex*-jGCaMP8m) showed, similar to *Camklla*⁺_{DPGi,Pr} neurons, activity during REMS and wake are increased compared with NREMS (Figures S5K–S5N). The REMS-promoting effect of *Nos1*⁺_{DPGi,Pr} neurons seemed even stronger than that of *Camklla*⁺_{DPGi,Pr} neurons (Figures 5P, 6C, 6D, 6J, and S4G), suggesting that *Nos1* labels the REMS-promoting neurons in DPGi/Pr with higher specificity or efficiency. We found that another subset of neurons that are distributed slightly ventrolateral to the *Nos1*⁺_{DPGi,Pr} neurons express corticotropin-releasing hormone (*Crh*) (*Crh*⁺_{DPGi,Pr} neurons; Figure S5A); these neurons were mostly glutamatergic based on the expression of *Vglut2* (Figure S5O) and partly overlapped with *Camklla*⁺_{DPGi,Pr} neurons but were distinct from *Nos1*⁺_{DPGi,Pr} neurons (Figure S5A). Consistent with a recent study showing that *Crh* neurons in the dorsal medulla promote REMS,⁴¹ chemogenetic activation of

Figure 4. *Crhbp*⁺_{SubLDT→BF} neurons promote NREMS

- (A) Retrograde AAV-carrying Cre-dependent flippase was microinjected into BF, and AAV-carrying flippase-dependent hM3Dq-YFP was microinjected into SubLDT of *Crhbp*-Cre mice.
- (B) hM3Dq-YFP and mCherry were detected in SubLDT and BF, respectively. Note that mCherry, expressed independently of Cre, served as an infection marker of AAV-encoding Cre-dependent flippase (scale bar: 1 mm; green, hM3Dq-YFP; magenta, ChAT; yellow, mCherry).
- (C) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and hypnogram following administration of saline or CNO (1.5 mg/kg).
- (D) Comparison of the effect of saline and CNO (1.5 mg/kg) administered at ZT5 on the amount of each vigilance state. Horizontal axes indicate time after drug administration. Data represent mean ± SEM. *n* = 5 mice/group. **p* < 0.05; **,††*p* < 0.01; ***,†††,††††*p* < 0.001 (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).
- (E) Retrograde AAV-encoding Cre-dependent flippase was microinjected into BF, and AAV-encoding flippase-dependent jGCaMP7s was microinjected into SubLDT of *Crhbp*-Cre mice. The optic fiber was inserted above the SubLDT.
- (F) jGCaMP7s and mCherry were detected in SubLDT and BF, respectively. mCherry, expressed independently of Cre, served as an infection marker of the AAV-carrying Cre-dependent flippase (scale bar: 1 mm; green, jGCaMP7s; magenta, ChAT; yellow, mCherry; blue, DAPI).
- (G) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and photometry signal.
- (H) Comparison of fluorescent signals among each vigilance state. Data represent mean ± SEM. *n* = 8 mice. **††*p* < 0.01, (one-way ANOVA followed by Tukey's test, † indicates significant difference among vigilance states in one-way ANOVA, and * indicates significance in Tukey's test).
- (I) Photometry signals aligned to vigilance state transitions. Data represent mean ± SEM. Wake to NREMS, *n* = 51 episodes obtained from 4 mice; NREMS to wake, *n* = 17 episodes obtained from 4 mice; NREMS to REMS, *n* = 34 episodes obtained from 4 mice; REMS to wake, *n* = 17 episodes obtained from 4 mice. ****p* < 0.001 (paired Student's *t* test).
- (J) Co-expression analysis of hM3Dq-YFP with *Vglut2* (ISH) or *Vgat* (ISH). Data represent mean ± SEM. *n* = 6 sections, obtained from 2 mice. (Scale bar: 100 μm; yellow, hM3Dq-YFP; magenta, *Vglut2* [ISH]; cyan, *Vgat* [ISH]).
- (K and L) AAV-encoding Cre-dependent hM3Dq-mCherry were microinjected into SubLDT of *Crhbp*-Cre mice (scale bar: 1 mm; white, hM3Dq-mCherry; blue, DAPI).
- (M) Time spent in each vigilance state following chemogenetic activation of *Crhbp*⁺_{SubLDT} neurons. Saline or CNO (1.5 mg/kg) was administered in the light phase (ZT5). Horizontal axes indicate time after drug administration. Data represent mean ± SEM. *n* = 5 mice/group. †, **p* < 0.05; ***p* < 0.01; ***,†††*p* < 0.001 (two-way repeated-measures ANOVA followed by Bonferroni's test; † indicates significant interaction between intervention and time, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).

See also Figure S2.



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Crh⁺_{DPGi,Pr} neurons (*Crh*-Cre mice, DPGi/Pr: AAV.2-*hSyn-DIO-hM3Dq-mCherry*) increased REMS (Figures S5P–S5U), although the effect seemed milder compared with the activation of *Nos1*⁺_{DPGi,Pr} neurons or *Camklla*⁺_{DPGi,Pr} neurons (Figures 5P, 6C, 6D, S4G, and S5P–S5U).

Notably, chemogenetically activating *Nos1*⁺_{DPGi,Pr} neurons or *Camklla*⁺_{DPGi,Pr} neurons caused direct transitions from wake to REMS (Figures 6H and 6I). In addition, in cases where NREMS preceded REMS, the duration of the preceding NREMS episodes was extremely short (Figure 6J). Transitions from wake to REMS occur only in narcoleptic patients or animals defective of the orexin/hypocretin system,^{42–44} indicating that such transitions are strongly inhibited by the orexin/hypocretin system. Thus, the effect of *Nos1*⁺_{DPGi,Pr} neurons seems to overwhelm the gating of REMS by orexin/hypocretin. Activation of *Crhbp*⁺_{SubLDT→Gi} neurons or *Crh*⁺_{DPGi,Pr} neurons did not cause such transitions (Figure 6J), further depicting the robust effects of *Nos1*⁺_{DPGi,Pr} neurons in REMS induction.

We next examined downstream targets of *Nos1*⁺_{DPGi,Pr} neurons. We injected AAV-carrying Cre-dependent tdTomato into DPGi/Pr of *Nos1*-Cre mice (Figure 6K). Within the pons, tdTomato-labeled axons were detected mostly in SubLDT (Figure 6L), consistent with the model in which reciprocal projections between the SubLDT and DPGi/Pr are crucial

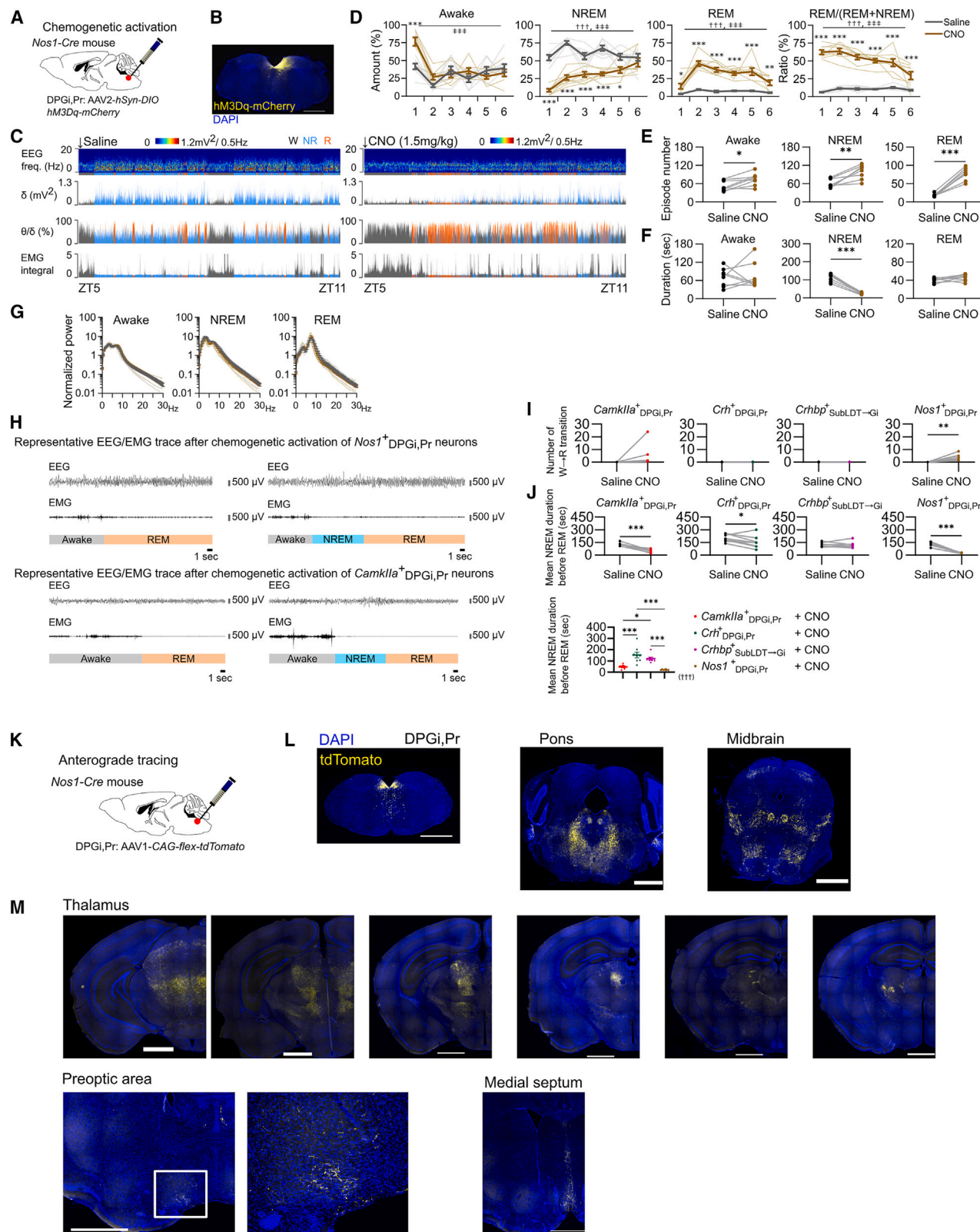
for REMS induction. In addition, tdTomato-labeled axons were detected in wide areas of the thalamus, including the parafascicular nucleus, mediodorsal thalamus, central medial thalamic nucleus, paracentral thalamic nucleus, central lateral thalamic nucleus, ventral posterolateral thalamic nucleus, and ventral posteromedial thalamic nucleus. tdTomato signals were also detected in the oculomotor nucleus, preoptic area of the hypothalamus, and medial septum (Figures 6L and 6M). Taken together, these findings identify a REMS-inducing circuit consisting of a reciprocal loop between the pons and medulla and projections from the medulla to the forebrain.

Selective loss of CRHBP-IR neurons in PD patients with RBD

Sleep-related symptoms of PD, including deficits in muscle atonia during REMS and reduction of REMS and slow-wave sleep, well-match the phenotypes of mice with *Crhbp*⁺_{S,L,V} neuron ablation (Figure 1). Having established *Crhbp* as a marker for sleep-regulating neurons, we examined whether these neurons are affected in PD patients with RBD. We generated a polyclonal antibody for human CRHBP (amino acid residues [aa] 98–136). We then conducted CRHBP immunohistochemistry of post-mortem brains derived from PD patients with RBD and control patients without

Figure 5. *Camklla*⁺_{DPGi,Pr} neurons promote REMS via projection to SubLDT

- (A) Mixture of AAV-carrying Cre-dependent TVA-mCherry and AAV-carrying Cre-dependent RG was microinjected into SubLDT of *Crhbp*-Cre mice. At 3 weeks after AAV microinjection, RV-SADΔG-GFP (EnvA) was microinjected into Gi.
- (B) Starter cells (double-positive for TVA-mCherry and RV-GFP) were detected in SubLDT of *Crhbp*-Cre mice. (Scale bar: 1 mm; green, RV-GFP; magenta, TVA-mCherry; blue, DAPI).
- (C) Neurons projecting to *Crhbp*⁺_{SubLDT→Gi} neurons (single-positive for RV-GFP) were detected in dDpMe, LPGi, DPGi, and Pr. (Scale bar: 1 mm; green, RV-GFP; blue, DAPI).
- (D) DPGi/Pr neurons projecting to *Crhbp*⁺_{SubLDT→Gi} neurons (single-positive for RV-GFP) express either *Vglut2* (left) or *Vgat* (middle) (scale bar: 10 μm; green, GFP; magenta, *Vglut2*; orange, *Vgat*), and a part of these neurons can be labeled with the AAV-*Camklla* promoter (right) (scale bar: 10 μm; green, RV-GFP; magenta, mCherry fluorescence derived from AAV-*Camklla*-mCherry-Cre; blue, DAPI).
- (E) AAV-encoding GFP was microinjected into DPGi/Pr of WT mouse. GFP-labeled axons were detected in SubLDT. (Scale bar: 1 mm; green, GFP; magenta, ChAT; blue, DAPI; 4V, fourth ventricle).
- (F and G) AAV-encoding SwiChR++-YFP was microinjected into DPGi/Pr, and mixture of AAV-carrying Cre-dependent DTA and AAV-carrying GFP-Cre was microinjected into left SubLDT of WT mouse. The optic fiber was inserted above right SubLDT. (Scale bar: 0.5 mm; green, SwiChR++-YFP; blue, NeuN).
- (H) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral with (right) or without (left) blue laser stimulation (462 nm, 1 s/20 min for 4 h, 25 mW).
- (I) Comparison of time spent in each vigilance state during the 4 h with or without laser stimulation. Data represent mean ± SEM, *n* = 6 mice, ***p* < 0.01 (unpaired *t* test).
- (J) Retrograde AAV-encoding Cre was microinjected into SubLDT, and AAV-encoding Cre-dependent hM3Dq-mCherry was microinjected into DPGi/Pr of WT mice, respectively, to express hM3Dq-mCherry in *Camklla*⁺_{DPGi,Pr→SubLDT} neurons.
- (K) Cre and hM3Dq-mCherry were detected in SubLDT and DPGi/Pr, respectively. (Scale bar: 1 mm; green, Cre; yellow, hM3Dq-mCherry; blue, DAPI).
- (L) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral following administration of saline or CNO (1.5 mg/kg).
- (M) Comparison of time spent in each vigilance state during the 3 h following administration of saline or CNO (1.5 mg/kg) at ZT5. Data represent mean ± SEM, *n* = 5 mice, **p* < 0.05 (unpaired *t* test).
- (N and O) AAV-encoding hM3Dq-mCherry or hM4Di-mCherry was microinjected into DPGi/Pr of WT mice to express hM3Dq or hM4Di in *Camklla*⁺_{DPGi,Pr} neurons. (Scale bar: 1 mm; red, hM3Dq-mCherry; yellow, hM4Di-mCherry; blue, DAPI).
- (P) Comparison of time spent in each vigilance state during the 6 h following administration of saline or CNO (1.5 mg/kg) at ZT5. Horizontal axes indicate time after drug administration. Data represent mean ± SEM. *n* = 6 mice in hM3Dq and GFP group, *n* = 4 mice in hM4Di group. *, †*p* < 0.05; **, ††, ‡‡*p* < 0.01; ***, †††, ‡‡‡*p* < 0.001 (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).
- (Q and R) AAV-encoding GCaMP6s were microinjected into DPGi/Pr of WT mice to express GCaMP6s in *Camklla*⁺_{DPGi,Pr} neurons. The optic fiber was inserted above the DPGi/Pr.
- (S) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and photometry signal.
- (T) Comparison of fluorescent signals among each vigilance state. Data represent mean ± SEM. *n* = 8 mice. **p* < 0.05, ††*p* < 0.01 (one-way ANOVA followed by Tukey's test, † indicates significant difference in one-way ANOVA, and * indicates significance in Tukey's test).
- See also Figures S3 and S4.



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neurodegeneration (Table S1). The mouse SubLDT is characterized by its location close to the periaqueductal gray and LC and the sparse distribution of cholinergic and noradrenergic neurons (Figure S6A). To identify the human SubLDT, we performed immunohistochemical analyses for dopamine beta-hydroxylase (DBH) and choline acetyltransferase (ChAT). Ventral to the most rostral portion of LC, we observed an area in which noradrenergic and cholinergic neurons were sparsely distributed, similar to the mouse SubLDT (Figure S6B). Moreover, CRHBP-immunoreactive (CRHBP-IR) neurons were detected in this area. The number of CRHBP-IR neurons in this area was reduced in PD patients with RBD compared with control patients (Figures 7A, 7B, S6B, and S6C). By contrast, the number of cholinergic neurons was not significantly reduced, suggesting specific loss of CRHBP-IR neurons (Figures 7A and 7B). Results were similar when we used a commercially available human CRHBP (aa 25–322) antibody (Figures 7C and 7D). The number of CRHBP-IR nor cholinergic neurons correlated with post-mortem intervals at fixation nor with durations of the fixation period (Figure S6J). We next tested the specificity of CRHBP antibodies by 2 methods, i.e., western blotting and immunocytochemistry. When human brain homogenate was subjected to western blotting using CRHBP (aa 98–136) antibody, 2 bands were detected: 1 near the expected size of CRHBP (37 kDa) and the other around 50 kDa (Figure S6D). Both bands were also detected when cultured human cells were subjected to western blotting (Figure S6E). Overexpression or small interference RNA of CRHBP increased or decreased the intensity, respectively, of the 37-kDa band but not the 50-kDa band (Figure S6E), suggesting that the 50-kDa band is due to some non-specific action of the antibody. Western blotting using CRHBP (aa 25–322) antibody against human brain homogenate and cultured human cells also detected the 37- and 50-kDa bands (Figures S6F and S6G). Considering that polyclonal antibodies designed for immunohistochemistry are frequently not optimal for western blotting, we also conducted immunocytochemistry. Cultured human cells expressing either full-length CRHBP, CRHBP lacking the recognition site of CRHBP (aa

98–136) antibody (CRHBP Δaa 98–136), or CRHBP lacking the recognition site of CRHBP (aa 25–322) antibody (CRHBP Δaa 25–322) were generated and subjected to immunocytochemistry. As a result, immunocytochemistry using CRHBP (aa 98–136) antibody produced clear signals in cells overexpressing full-length CRHBP but not CRHBP Δaa 98–136 or CRHBP Δaa 25–322, whereas immunocytochemistry using CRHBP (aa 25–322) antibody produced clear signals in cells overexpressing full-length CRHBP or CRHBP Δaa 98–136 but not CRHBP Δaa 25–322 (Figure S6H). In case of cells expressing full-length CRHBP, signals derived from the 2 antibodies largely overlapped with each other (Figure S6H). To test specificity of the antibodies by immunohistochemistry, we injected AAV that co-express human *Crhbp* and *GFP* into the mouse thalamus, which does not express endogenous *Crhbp*, and conducted immunohistochemistry. As a result, for both antibodies, signals were only detected in AAV-injected mice and largely overlapped with GFP-expressing cells (Figure S6I).

We also detected the accumulation of phosphorylated S129 α -synuclein, a component of the Lewy body, in CRHBP-IR neurons of PD patients (Figure 7E), suggesting that *Crhbp*^{SubLDT} neurons are susceptible to α -synuclein aggregation, which might account for the selective loss of these neurons in PD patients. These results strongly suggest that impairment of *Crhbp*^{SubLDT} neurons contributes to RBD and other sleep-related symptoms in PD. Finally, we examined whether artificial activation of *Crhbp*^{SubLDT→Gi} neurons can improve sleep deficits in a mouse model of PD. We utilized the BAC-SNCA^{A53T} mouse, which is a bacterial artificial chromosome transgenic mouse that harbors the α -synuclein gene (SNCA) with 3 point mutations, resulting in progressive accumulation of aggregated α -synuclein while maintaining the native expression pattern of SNCA.⁴⁵ In *Crhbp-Cre*; BAC-SNCA^{A53T} double transgenic mice, RBD-like phenotypes were not detected at the age of the experiment (5 months), consistent with a previous study,⁴⁶ but both REMS and NREMS were reduced, the episode number of REMS was decreased, and the episode duration of wake was prolonged

Figure 6. *Nos1*^{DPGi,Pr} neurons strongly promote REMS and project to SubLDT and wide areas of the thalamus and medial septum

(A and B) AAV-carrying Cre-dependent hM3Dq-mCherry was microinjected into DPGi/Pr of *Nos1-Cre* mice (scale bar: 1 mm; yellow, hM3Dq-mCherry; blue, DAPI).

(C) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral following administration of saline or CNO (1.5 mg/kg). (D) Comparison of time spent in each vigilance state during the 6 h following administration of saline or CNO (1.5 mg/kg) at ZT5. Horizontal axes indicate time after drug administration. Data represent mean $\pm$ SEM. $n = 8$ mice. $^{**}p < 0.01$; $^{***}, \dagger, \ddagger, \dagger\dagger, \dagger\dagger\dagger p < 0.001$ (two-way repeated-measures ANOVA followed by Bonferroni's test; $\dagger$ and $\ddagger$ indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).

(E–G) Sleep architecture (E and F) and EEG power spectra (G) during the 3 h following saline or CNO (1.5 mg/kg) administration at ZT5. $n = 8$ mice. $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (paired Student's *t* test). Data represent mean $\pm$ SEM.

(H) Representative EEG/EMG trace during entry to REMS following chemogenetic activation of *CamkIIa*^{DPGi,Pr} neurons or *Nos1*^{DPGi,Pr} neurons, respectively. (I and J) Number of REMS episodes directly entered from wakefulness (I) and mean NREMS duration before REMS episodes (J) following chemogenetic activation of the REMS-promoting populations, namely *CamkIIa*^{DPGi,Pr} ($n = 6$ mice), *Nos1*^{DPGi,Pr} ($n = 8$ mice) *Crh*^{DPGi,Pr} ($n = 8$ mice), and *Crhbp*^{SubLDT→Gi} neurons ($n = 8$ mice). $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (paired Student's *t* test) for comparison of saline vs. CNO. $^*p < 0.05$; $^{***}, \dagger, \ddagger, \dagger\dagger, \dagger\dagger\dagger p < 0.001$ (one-way ANOVA followed by Tukey's test, $\dagger$ indicates significant difference in one-way ANOVA, and * indicates significance in Tukey's test) for comparison of the effect of CNO among *CamkIIa*^{DPGi,Pr}, *Crh*^{DPGi,Pr}, *Crhbp*^{SubLDT→Gi} neurons, or *Nos1*^{DPGi,Pr} neurons.

(K and L) AAV-carrying Cre-dependent tdTomato was microinjected into DPGi/Pr of *Nos1-Cre* mice.

(M) tdTomato signals were detected in the pons, the midbrain, including the oculomotor nucleus, wide area of the thalamus, preoptic area, and medial septum (scale bar: 1 mm; yellow, tdTomato; blue, DAPI).

See also Figure S5.

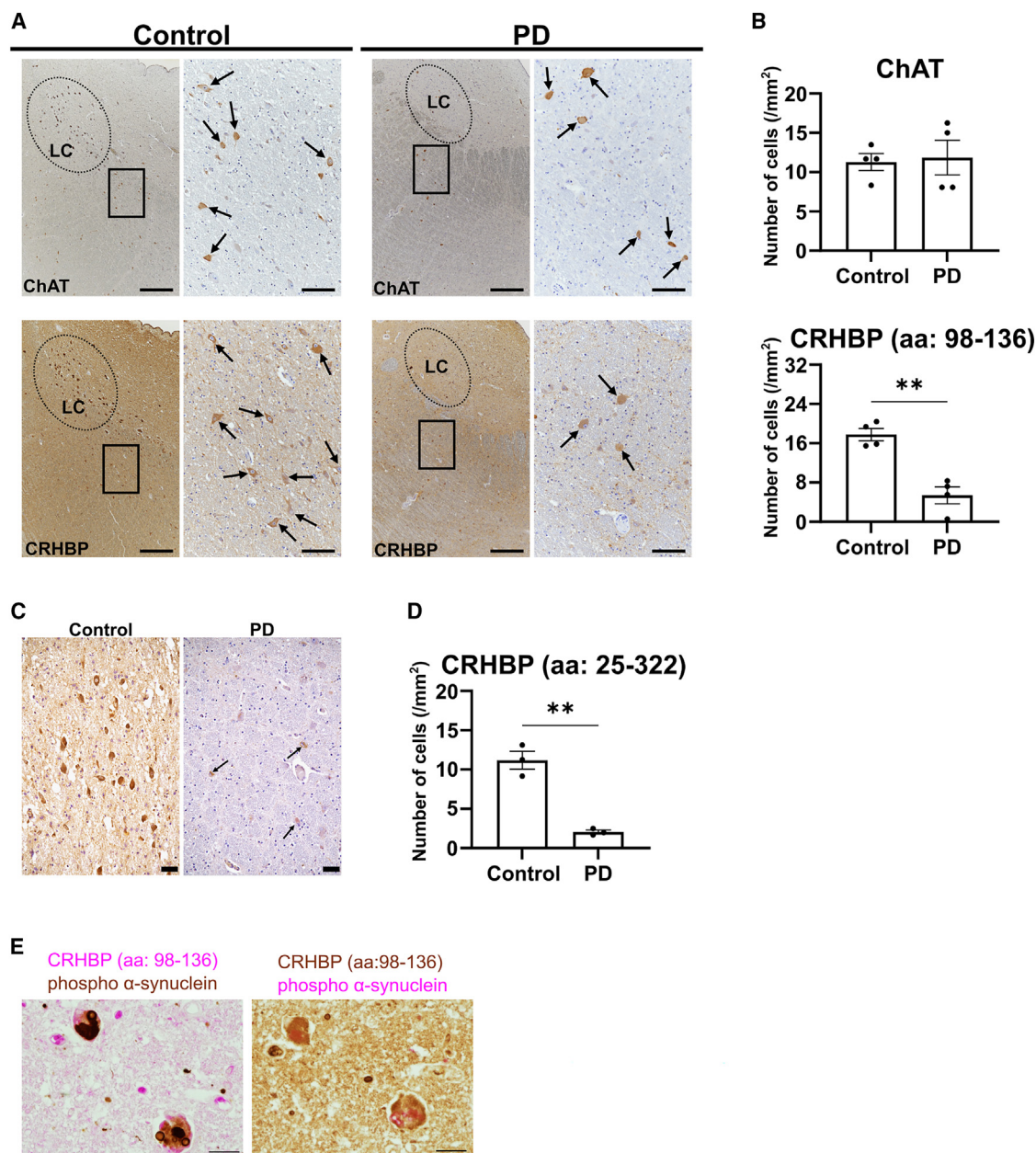


Figure 7. Selective loss of CRHBP-IR neurons in human PD patients with RBD

(A) Representative images of immunohistochemistry for ChAT (upper) or CRHBP (lower) in dorsal portions of the pons derived from control patients without neurodegenerative diseases (left) or PD patients with RBD (right). ChAT- or CRHBP-IR neurons are visible ventral to the locus coeruleus (LC). Scale bar: 500 μ m (low power field) and 100 μ m (high power field).

(B) Quantitative analyses of ChAT- or CRHBP-IR neurons in the human SubLDT. Data represent mean $\pm$ SEM, $n = 4$ samples, $**p < 0.01$ (bootstrap).

(C) Representative images of immunohistochemistry for CRHBP using a different antibody (anti-CRHBP (aa 25–322)). Scale bar: 100 μ m.

(D) Quantitative analyses of CRHBP-IR neurons in SubLDT conducted using the anti-CRHBP (aa 25–322) antibody. Data represent mean $\pm$ SEM, $n = 3$ samples, $**p < 0.01$ (bootstrap).

(E) CRHBP-IR neurons harboring Lewy bodies are visible in the human SubLDT. Scale bar; 50 μ m. Detailed information about clinical diagnosis and tissue preparation is shown in Table S1.

See also Figures S6 and S7.

compared with control *Crhbp-Cre* mice (Figures S7B–S7D). All these sleep deficits recovered to a level comparable to control mice upon chemogenetic activation of *Crhbp*⁺_{SubLDT→Gi} neu-

rons (Figures S7A–S7D), suggesting that artificial activation of *Crhbp*⁺_{SubLDT→Gi} neurons might ameliorate sleep deficits in PD.

DISCUSSION

Although it is widely accepted that the SLD plays a critical role in REMS regulation, the entity of neurons underlying this function has remained unidentified due to the heterogeneity of neurons. For example, simply activating SLD glutamatergic neurons increases wakefulness.¹⁴ In this study, the combination of a molecular marker and projection target analysis enabled us to identify *Crhbp*⁺_{SubLDT→Gi} neurons as the REMS-inducing neurons, which further allowed us to identify a REMS-inducing circuit including DPGi/Pr and SubLDT. In addition to *Crhbp*⁺_{SubLDT→Gi} neurons, activation of *Camklla*⁺_{DPGi,Pr} neurons strongly induced REMS in both light and dark phases. The *Camklla*⁺_{DPGi,Pr} neurons comprise glutamatergic and inhibitory neurons, and both subtypes are crucial for REMS. The *Camklla*⁺_{DPGi,Pr} neurons likely activate *Crhbp*⁺_{SubLDT→Gi} neurons and other as-yet unidentified REMS-promoting neurons in SubLDT, possibly via the release of some excitatory neuromodulators. We identified *Nos1* as a candidate molecular marker for the REMS-inducing *Camklla*⁺_{DPGi,Pr} neurons. These neurons heavily project to SubLDT, and their activation robustly increases REMS. *Crhbp*⁺_{SubLDT→Gi} neurons project to DPGi/Pr, suggesting a reciprocal positive interaction within the REMS-promoting neurons in SubLDT and DPGi/Pr. This positive interaction might allow rapid and reliable activation of REMS-promoting neurons in both areas during transitions from NREMS to REMS and subsequent persistent activity throughout the REMS episode. We also found that *Crhbp*⁺_{SubLDT→Gi} neurons receive projection from neurons in the dDpMe/ventrolateral periaqueductal gray (vlPAG), which contain GABAergic neurons that strongly inhibit REMS.^{12,18} This input from dDpMe/vlPAG might be crucial for the stabilization of wake and/or NREMS.

During transitions from NREMS to REMS, the cortical activity in wide areas changes rapidly, i.e., low-frequency oscillatory activities are suppressed, whereas average firing rates of pyramidal cells increase⁴⁷ via yet unclear mechanisms. Contrary to the classic notion that SLD neurons projecting to the thalamus are crucial for cortical activation during REMS, such thalamus-projecting neurons in the SLD are not activated during REMS hypersomnia.¹⁰ Moreover, we found that activation of *Crhbp*⁺_{SubLDT→Gi} neurons, i.e., a descending population of SLD neurons is sufficient to induce features of cortical activity characteristic of REMS. While neurons that control muscle atonia are located in the ventral medulla,^{11,21} axonal terminals of *Crhbp*⁺_{SubLDT→Gi} neurons are widely distributed in the medulla. Some of the target neurons in the medulla might directly or indirectly activate the forebrain via unknown ascending circuits. Indeed, we found that *Nos1*⁺_{DPGi,Pr} neurons densely project to wide areas of the thalamus in addition to SubLDT. Within the thalamus, the centromedial thalamus regulates brain-wide cortical activity,⁴⁸ and neurons in the intralaminar area are activated after REMS hypersomnia,⁶ both of which were found to receive massive innervation from *Nos1*⁺_{DPGi,Pr} neurons. Thus, *Nos1*⁺_{DPGi,Pr} neurons might regulate cortical activation during REMS. In addition, *Nos1*⁺_{DPGi,Pr} neurons also project to the medial septum, and this projection might regulate theta oscillations in the hippocampus during REMS. Considering the dense

projection of *Nos1*⁺_{DPGi,Pr} neurons to the thalamus and the medial septum and that of *Crhbp*⁺_{SubLDT→Gi} neurons to the ventral medulla, and considering previous reports showing that glutamatergic neurons projecting from SLD to the ventral medulla regulate muscle atonia,^{9–11} each neuronal population might be particularly responsible for the modulation of forebrain activity and induction of muscle atonia during REMS, respectively, and the positive interaction between the two populations might ensure the simultaneous induction of the two events.

Finally, we found that CRHBP-IR neurons within the human SubLDT are reduced in PD patients with RBD. According to Braak's model,^{23,49} the spread of α -synuclein begins in areas close to the periphery, such as the olfactory bulb from the olfactory epithelium or lower brainstem from the gut, and then widely proceeds to other brain areas over time. Considering that RBD is an early prodromal symptom of PD, *Crhbp*⁺_{SubLDT} neurons might act as an entry point for subsequent brain-wide propagation of α -synuclein. Moreover, considering that sleep contributes to the clearance of waste⁵⁰ and that sleep loss leads to increased α -synuclein in the cerebrospinal fluid,⁵¹ loss of *Crhbp*⁺_{SubLDT} neurons might contribute to the accumulation of α -synuclein and subsequent development and/or progression of PD. We also demonstrated that artificial activation of *Crhbp*⁺_{SubLDT→Gi} neurons rescues sleep deficits in a mouse model of PD. Taken together, *Crhbp*⁺_{SubLDT} neurons might be a target for ameliorating PD progression.

Although 2 states of sleep have been observed in various animals,⁵² the precise evolutionary origin of REMS and NREMS is yet unknown. Fate mapping of the *Crhbp*⁺_{SubLDT} and *Nos1*⁺_{DPGi,Pr} neurons and analyses of neurons homologous to these neurons in animal species in which these 2 sleep stages are not obvious might provide key insights into the evolution and function of sleep.

Limitations of the study

The results of human studies do not directly show that loss of CRHBP-IR neurons underlies the sleep deficits in PD patients. It also remains unknown why CRHBP-IR neurons are vulnerable in PD patients. Future studies combining human studies and PD mouse model studies are essential to clarify these points. In addition, it is important to confirm the loss of *Crhbp*⁺_{SubLDT} neurons in PD patients by direct measurement of *Crhbp* mRNA, such as by *in situ* hybridization, etc., as such approach exerts higher specificity compared with immunohistochemistry. We showed that chemogenetic activation of *Crhbp*⁺_{SubLDT→Gi} neurons in a PD mouse model rescues the sleep abnormalities observed in these mice. In the future, it is important to examine whether these findings provide clues to developing methods for improving sleep deficits that accompany PD.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Yu Hayashi (yuhayashi@ecc.u-tokyo.ac.jp).

Materials availability

Materials generated in this study will be available upon reasonable request.

Data and code availability

All raw data analyzed in this paper have been deposited at Mendeley (Mendeley Data: <https://doi.org/10.17632/46jxf7z75.1>) and are publicly available as of the date of publication. Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon reasonable request. This paper does not report original code.

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AUTHOR CONTRIBUTIONS

Conceptualization, M. Kashiwagi and Y.H.; methodology, M. Kashiwagi, G.B., M. Kanuka, Y.C.S., K.B., S.M., Y.O., M.S., M.I., H.Y., R.T., T.S., Y.N., M.W., H.M., and Y.H.; investigation, M. Kashiwagi, G.B., M. Kanuka, Y.A., K.T., C.T., M.T., Y.N., and Y.H.; project administration, Y.H.; supervision, M.Y., K.S., H.M., and Y.H.; writing – original draft, M. Kashiwagi and Y.H.; writing – review and editing, G.B., H.M., and Y.H.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Original data	This paper; Mendeley Data	https://doi.org/10.17632/46jxf7z75.1
Antibodies		
Rabbit anti-RFP	MBL	PM005, RRID:AB_591279
Goat anti-tdTomato	SICGEN	AB8181-200 RRID:AB_2722750
Rabbit anti-cfos	Millipore	ABE457 RRID:AB_2631318
Rat anti-GFP	Nacalai tesque	GF090R RRID:AB_2314545
Rabbit anti-denatured GFP	Santa Cruz	SC8334 RRID:AB_641123
Sheep anti-DIG-AP	Roche	11093274910 RRID:AB_514497
Goat anti-ChAT	Millipore	AB144P RRID:AB_2079751
Rabbit anti-Cre	Novagen	69050-3 RRID:AB_2314229
Mouse anti-NeuN	Millipore	MAB377 RRID:AB_2298772
Mouse anti-TH	Sigma-Aldrich	T2928; RRID:AB_477569
Rabbit anti-CRHBP (human); aa98-136	This study	N/A
Goat anti-CRHBP (human); aa25-322	R&D Systems	AF2796; RRID:AB_2084422
Rabbit anti-ChAT	Millipore	AB143; RRID:AB_11212924
Chemicals, peptides, and recombinant proteins		
Clozapine-N-oxide (CNO)	Abcam	ab141704
Clozapine-N-oxide (CNO)	Enzo Life Sciences	BML-NS105-0005
RNAscope Manual Fluorescent Multiplex kit V2	Advanced Cell Diagnostics	323100
Experimental models: Cell lines		
293T human kidney cells	Riken BRC	RCB2202; RRID:CVCL_0063
293A human kidney cells	Thermo Fisher Scientific	R70507; RRID:CVCL_6910
Experimental models: Organisms/strains		
Mouse: C57BL/6J	Charles River	N/A
Mouse: <i>Crhbp-2A-Cre</i>	This study	N/A
Mouse: <i>Rosa26-CAG-loxP-STOP-loxP-Rpl10a-GFP</i>	Jackson	022367
Mouse: <i>Vglut2-flox</i>	Jackson	012898
Mouse: <i>Vglut2-Cre</i>	Jackson	016963
Mouse: <i>Vgat-Cre</i>	Jackson	016962
Mouse: <i>Nos1-Cre</i>	Laboratory Animal Resource Bank, National Institutes of Biomedical Innovation, Health and Nutrition	Bio334
Mouse: <i>Crh-Cre</i>	Jackson	012704
Mouse: BAC-SNCA ^{A53T}	Yamakado et al. ⁴⁵	N/A
Recombinant DNA		
AAV-hSyn-DO-mCherry-DIO-flippase	This study	N/A
AAV-hSyn-fDIO-hM3Dq-YFP	This study	N/A
AAV-hSyn-fDIO-jGCaMP7s	This study	N/A
AAV-CamkIIa-DIO-hM3Dq-mCherry	This study	N/A
AAV-CamkIIa-DIO-hM4Di-mCherry	This study	N/A
AAV-hSyn-GFP-T2A-CRHBP	This study	N/A
AAV-EF1a-flex-TVA-mCherry	Addgene	69618
AAV-CAG-flex-RG	Addgene	48333

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
RV-SADΔG-GFP-ERT2-Cre-ERT2	Addgene	32649
AAV-CamkIIa-hM3Dq-mCherry	UNC Chapel Hill Vector Core	N/A
AAV-CamkIIa-hM4Di-mCherry	UNC Chapel Hill Vector Core	N/A
AAV-CamkIIa-GFP	UNC Chapel Hill Vector Core	N/A
AAV-CamkIIa-mCherry-Cre	UNC Chapel Hill Vector Core	N/A
AAV-Ef1a-mCherry-flex-DTA	UNC Chapel Hill Vector Core	N/A
AAV-hSyn-GFP-Cre	UNC Chapel Hill Vector Core	N/A
AAV-CamkIIa-SwiChR++-YFP	UNC Chapel Hill Vector Core	N/A
AAV-hSyn-DIO-hM3Dq-mCherry	Addgene	44361
AAV-hSyn-DIO-hM4Di-mCherry	Addgene	44362
AAV-hSyn-flex-tdTomato-T2A-synaptophysin-GFP	Salk vector core	N/A
AAV-mSyn-BFP-Cre	Addgene	51507
AAV-CAG-flex-tdTomato	Addgene	51503
AAV-hSyn-DIO-EGFP	Addgene	50457
AAV-hSyn-flex-jGCaMP8m	Addgene	162378
AAV-CamkIIa-GCaMP6s	University of Pennsylvania Vector Core	N/A
Software and algorithms		
Sleep Sign	Kissei Comtec	N/A
Vital recorder	Kissei Comtec	N/A
Prism	GraphPad	N/A
Doric Neuroscience studio	Doric Lenses	N/A
ImageJ	NIH	https://imagej.nih.gov/ij/
R	Foundation for Statistical Computing	https://www.r-project.org/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**Animals**

All animal experiments were approved by the institutional animal care and use committee of the University of Tsukuba or University of Tokyo, respectively. All animals were maintained according to the institutional guidelines of the Laboratory of Animal Resource Center of the University of Tsukuba or the institutional guidelines of University of Tokyo, respectively.

Following our previous microarray analysis (NCBI Gene Expression Omnibus: GSE143280),¹³ we ranked candidate genes based on the following criteria: “target to control signal ratio larger than 1.2 were selected and ranked according to the F value in one-way ANOVA (F(2,6))”. With the full list, starting from the top (*Neurotensin* (*Nts*)), we conducted either IHC or ISH of the candidate genes and searched for genes whose expression show selectivity to SubLDT and do not largely overlap with cholinergic or noradrenergic neurons. Based on these processes, *Corticotropin releasing hormone binding protein* (*Crhbp*) was the first to satisfy the criteria. (Of note, for most candidate genes, ISH or IHC did not provide an obvious signal, and thus we could not proceed further.) *Crhbp-Cre* mice, in which *P2A-Cre-pA* was inserted immediately before the stop codon of *Crhbp*, were generated using CRISPR/Cas9 technology at the Laboratory of Animal Resource Center of the University of Tsukuba as previously described.⁵³ The targeted sequence for CRISPR/Cas9 was 5'-CTGTTTGTCTAGTCTGTGAATGG-3'. A donor plasmid containing *P2A-Cre-pA* flanked by 5'- and 3'-homology arms (1.4 kbp each) and plasmid pX330⁵⁴ carrying Cas9 and the gRNA were microinjected into C57BL/6J zygotes. All the F0 progeny were screened by polymerase chain reaction (PCR) with the primer pair for *Cre*. For *Cre*-positive progeny, insertion of the transgene into the *Crhbp* locus was tested by PCR using the following primer pairs: *upstream of the 5' arm of the donor plasmid - Cre*; 5'-ACAGTGGTGGCCTGTCTCAG -3' & 5'-CGACCGTAATGCAGGCAAA -3', and *pA - downstream of the 3' arm of the donor plasmid*; 5'-TATGCTGGCTGCCATGAACAAAG -3' & 5'-CGCAGGAGAGGCAGGCTTAT -3'. We also tested the lack of random insertions of the injected plasmids by a primer pair that amplifies a part of the vector backbone: *Ampicillin-resistant*; 5'-TTCTGCTATGTGGCGCGGTA-3' & 5'-GCTTACCATCTGGCCCCAGT-3'. Subsequently, we confirmed that only a single copy of *Cre* was inserted by quantitative PCR by comparing the results with those of *Ptf1a-Cre* mice,⁵⁵ in which a single copy of *Cre* insertion was previously confirmed. A line exhibiting germline transmission of the transgene, confirmed by natural mating with WT C57BL/6J mice, was established as the *Crhbp-Cre* strain. Finally, the expected transgene insertion by PCR and the single-copy insertion of *Cre* described above were confirmed again in the F1 generation.

All strains were maintained in a C57BL/6J background. Male and female mice at least 8 weeks old were used for analysis. The influence of sex was not addressed in this study. Mouse strains and the genotyping primer pairs used in this study

were: *Crhbp-Cre*; 5'- CACCCCAATTCAGCCCAGA -3' & 5'- TGTGTGCACGTGTTCAAGGC -3' & 5'- TATGCTGGCTGCCATGAA CAAAG -3', *Rosa26-CAG-loxP-STOP-loxP-Rpl10a-green fluorescent protein (GFP)*⁵⁶; 5'-AAGGGAGCTGCAGTGGAGTA-3' & 5'-CCGAAAATCTGTGGGAAGTC-3' & 5'- AAGATCCGCCACAACATCG-3', *TTCTCGTTGGGGTCTTTGCT* -3', *Vesicular glutamate transporter2 (Vglut2)-flox*⁵⁷; 5'- CCTGAGCGAAGGTGAGCTGAA-3' & 5'-CCAAATACCTGAAAGTTACTG-3' & 5'- TGGGGAAC CAGGTGCACAG -3'. *Vglut2-Cre*⁵⁸; 5'- ATCGACCGGTAATGCAGGCAA-3' & 5'- CGGTACCACCAAATCTTACGG-3' & 5'- CAT GGTC TGTTTGAATTCAG-3', *Cre* and *Vesicular gaba transporter (Vgat)-Cre*⁵⁸; 5'- GACGATGCAACGAGTGATGA-3' & 5'- AGCATT GCTGTCACTTGGTC-3', *Neuronal nitric oxide synthase 1 (Nos1)-Cre* (obtained from Laboratory Animal Resource Bank, National Institutes of Biomedical Innovation, Health and Nutrition. deposit# bio334); 5'- GAGGTCTTCTGAAGGACACAGATAC-3' & 5'- CAGGTTCTATCTCTTTGAGCAGTTC-3', *Corticotropin releasing hormone (Crh)-Cre*⁵⁹; 5'- CTTACACATTTCTGCTCTAGCC -3' & 5'- CACGACCAGGCTGCGGCTAAC -3' & 5'- CAATGTATCTTATCATGTCTGGATCC -3'; *BAC-SNCA*^{A53T transgenic}⁴⁵; 5'-ACTTGCT AGGCCACCTGAGA-3' & 5'-ATGCCAGGTGTTTGGAAAAG-3'. All mice were housed on a 12-h light-dark cycle with free access to food and water. All mice used in this study were group-housed (2-5 mice/cage) before surgery and single-housed after surgery.

Human participants

Clinical profiles of human autopsy cases (control patients without neurodegenerative diseases, n = 4; PD patients with RBD symptoms, n = 4) used for pathologic analyses are shown in Table S1. In all PD patients, RBD symptoms were found in the medical records. The age, sex, ancestry, and race/ethnicity of the participants are also shown in Table S1. The influence of sex, age, race or ethnicity on the pathological changes of CRHBP-positive cells is not known, and our study is not strong enough to address this point because only eight Japanese patients were analyzed. This study was approved by the Ethics Committee of Osaka University Hospital (no. 12148). Written informed consent was obtained from the family of the patients included in this study.

Cell lines

Human embryonic kidney (HEK) cells 293T and 293A, which were used for AAV packaging, Western blotting, and immunocytochemistry, were seeded on 100 mm plates and cultured in Dulbecco's Modified Eagle Medium with 10% fetal bovine serum and 1% penicillin-streptomycin in a 37°C, 5% CO₂ incubator.

METHOD DETAILS

Preparation of adeno-associated virus (AAV) and rabies virus (RV) (Mice)

To generate plasmid pAAV-*hSyn-DO-mCherry-DIO-flippase*, the *hM3Dq-mCherry* segment of pAAV-*hSyn-DIO-hM3Dq-mCherry*^{60,61} was replaced with an overlap extension (OE)-PCR fragment containing *DO-mCherry-DIO-flippase* mimicking the design of *DO-tdTomato-DIO-GFP* in the previous study.⁶² To generate plasmid pAAV-*hSyn-fDIO-hM3Dq-YFP*, the *YFP* segment of pAAV-*EF1a-fDIO-YFP*⁶³ was replaced with an OE-PCR fragment containing *hM3Dq-YFP*. Subsequently, the *EF1a* segment was replaced with a DNA fragment containing the *hSyn* promoter. To generate plasmid pAAV-*hSyn-fDIO-jGCaMP7s*, the *hM3Dq-YFP* segment of pAAV-*hSyn-fDIO-hM3Dq-YFP* was replaced with a DNA fragment containing *jGCaMP7s*.⁶⁴ To generate plasmid pAAV-*CamkIIa* (1.3kbp)-*DIO-hM3Dq-mCherry*, the *hSyn* segment of pAAV-*hSyn-DIO-hM3Dq-mCherry* was replaced with a DNA fragment containing *CamkIIa*. Subsequently, *WPRES* was removed by *Clal* digestion. To generate plasmid pAAV-*CamkIIa-DIO-hM4Di-mCherry*, the *hM3Dq-mCherry* segment of pAAV-*CamkIIa-DIO-hM3Dq-mCherry* was replaced with a DNA fragment containing *hM4Di-mCherry*. To generate plasmid pAAV-*hSyn-GFP-T2A-CRHBP*, the *GFP* segment of pAAV-*hSyn-GFP* (addgene plasmid # 50465) was replaced with a DNA fragment containing *GFP-T2A-CRHBP*.

AAV-*hSyn-fDIO-hM3Dq-YFP* (serotype 8), AAV-*hSyn-DO-mCherry-DIO-flippase* (serotype rg (retrograde)⁶⁵), AAV-*hSyn-fDIO-jGCaMP7s* (serotype 8), AAV-*CamkIIa-DIO-hM3Dq-mCherry* (serotype 8), AAV-*CamkIIa-DIO-hM4Di-mCherry* (serotype 8), and AAV-*hSyn-GFP-T2A-CRHBP* (serotype PHP.eB⁶⁶) were packaged as previously described.¹³ AAV-*EF1a-flex-TVA-mCherry* (serotype 10), AAV-*CAG-flex-RG* (serotype 10), and RV-SADΔG-GFP-ERT2-Cre-ERT2 (EnvA) were obtained as previously described.^{67–69} AAV-*CamkIIa-hM3Dq-mCherry* (serotype 8), AAV-*CamkIIa-hM4Di-mCherry* (serotype 8), AAV-*CamkIIa-GFP* (serotype 8), AAV-*CamkIIa-mCherry-Cre* (serotype 8), AAV-*Ef1a-mCherry-flex-DTA*⁷⁰ (serotype 1), AAV-*hSyn-GFP-Cre* (serotype 1), AAV-*CamkIIa-SwiChR++-YFP*⁷¹ (serotype 5), AAV-*hSyn-DIO-hM3Dq-mCherry*⁶¹ (serotype 2), AAV-*hSyn-DIO-hM4Di-mCherry*⁶¹ (serotype 8), AAV-*hSyn-flex-tdTomato-T2A-synaptophysin-GFP*⁷² (serotype 1), AAV-*mSyn-BFP-Cre*⁷³ (serotype rg), AAV-*CAG-flex-tdTomato*⁷² (serotype 1 or rg), AAV-*hSyn-DIO-EGFP* (serotype rg) (addgene plasmid #50457), AAV-*hSyn-flex-jGCaMP8m* (serotype 9) (addgene plasmid #162378), and AAV-*CamkIIa-GCaMP6s* (serotype 9) were purchased from UNC vector core (Chapel Hill, NC, USA), Salk vector core (La Jolla, CA, USA), the Penn vector core (Philadelphia, PA, USA), or Addgene (Watertown, MA, USA) respectively.

Surgery (Mice)

Mice were anesthetized with isoflurane and placed in a stereotaxic apparatus (Angle Two; Leica Microsystems, Wetzlar, Germany, Model 68045; RWD Life Science, Shenzhen, China, or Model 940; David KOPF instruments, Tujunga, CA, USA) and the heights of bregma and lambda were adjusted to be within 0.1 mm of each other. AAV vector for the SubLDT, LDT, and VT (0.4 μl), SubLDT (0.1–0.3 μl), Gi (0.05–0.1 μl), BF (0.5 μl), DPGi/Pr (0.1–0.5 μl), and SC (0.1 μl) was microinjected at a speed of 0.02–0.1 μl/min using a glass capillary and a syringe pump (210, KD Scientific, Holliston, MA, USA). Target locations were as follows (all ML (medio-lateral),

AP (antero-posterior), DV (dorso-ventral) locations are from bregma unless otherwise stated): SubLDT, LDT, and VT; (ML, ± 0.4 mm; AP, -5.2 mm; DV, -4.2 mm), SubLDT: (ML, ± 0.7 mm; AP, -5.2 mm; DV, -4.2 mm), Gi: (ML, ± 1.0 mm; AP, -6.8 mm; DV, -5.1 mm from brain surface), BF: (ML, ± 1.2 mm; AP, 0.3 mm; DV, -5.0 mm), DPGi/Pr: (ML, 0 or ± 0.2 mm; AP, -6.8 mm; DV, -5.3 mm), SC: (ML, ± 0.7 and ± 1.5 mm; AP, -4.1 mm; DV, -1.3 and -2.0 mm from brain surface), and thalamus: (ML, 1.3 mm; AP, -1.4 mm; DV, -4.0 mm). Except for the case of the thalamus, after the AAV microinjection, EEG (electroencephalogram) and EMG (electromyogram) electrodes were implanted. Electrodes (stainless steel recording screws) for EEG were implanted epidurally over the parietal area (1 mm anterior to lambda, 1 mm lateral to the midline) referenced to the cerebellum area. EMG electrodes were bilaterally implanted into the neck muscle. For optogenetics and fiber photometry, we implanted optic fiber (400 μ m NA 0.48; Doric Lenses, Quebec, Canada) above the SubLDT (ML, $\pm$ or ± 0.7 mm; AP, -5.3 mm; DV, -3.8 mm) or DPGi/Pr (ML, 0 mm; AP, -6.8 mm; DV, -5.0 mm), respectively.

Drug preparation (Mice)

Clozapine-N-oxide (ab141704; Abcam, Cambridge, UK, or BML-NS105-0005; Enzo Life Sciences, Farmingdale, NY, USA) was prepared at 5 mg/ml in saline and diluted with saline. For sleep recording experiments, CNO was administered at a dose of 1.5 mg/kg body weight. For c-fos staining, CNO was administered at a dose of 4.5 mg/kg body weight, and mice were killed 2 h after drug administration.

EEG/EMG recording and analysis (Mice)

Mice were allowed to recover for at least 2 weeks after surgery. The mice were then moved to a sleep recording chamber and connected to an amplifier for EEG/EMG recording. For acclimatization, the mice were administered vehicle (saline) by intraperitoneal injection at ZT5 for 5 days or at ZT17 for 3 days. EEG/EMG signals were recorded and analyzed for at least 6 h starting from within 1 min after the drug administration. In each mouse, EEG/EMG was recorded at least twice for vehicle and CNO, respectively, and the mean value for each condition was calculated. For DTA and cKO experiments, we recorded EEG/EMG for 24 h at 4 weeks after surgery. EEG/EMG signals were filtered (EEG: high pass 0.5–64 Hz [sampling rate: 128 Hz] / 0.5–250 Hz [sampling rate: 512 Hz], EMG: bandpass 10–40 Hz) and collected and digitized at a sampling rate of 128 or 512 Hz via VitalRecorder (Kissei Comtec, Matsumoto, Japan). EEG signals were subjected to fast Fourier transform and further analysis using SleepSign (Kissei Comtec, Matsumoto, Japan). The vigilance state in each epoch was manually classified as REMS, NREMS, or wake based on absolute delta (0.5–4 Hz) power, theta (6–10 Hz) power to delta power ratio, and the integral of EMG signals. We applied 4 s epochs (time window) throughout the study. If a single epoch contained multiple states, the state with the highest occupancy was assigned.

Fiber photometry (Mice)

For fiber photometry, mice were allowed to recover for at least 3 weeks after surgery. The mice were then moved to a recording chamber and connected to an amplifier for EEG/EMG recording. For acclimatization, the mice were connected to an optic patch code (Doric Lenses, Quebec, Canada) for at least 3 days. Fiber photometry recordings were conducted with a Doric fiber photometry system (Doric Lenses, Quebec, Canada) following the manufacturer's protocol. We used 2 different excitations; 465 nm to record GCaMP-dependent and 405 nm to record GCaMP-independent signals that allow us to monitor motion artifacts. Signals were passed through an emission filter (500 – 550 nm) and detected at a dorc fluorescence detector (Doric Lenses, Quebec, Canada). Recordings were performed with Neuroscience studio (Doric Lenses, Quebec, Canada) at lock-in amplification, and signals were collected and digitalized at a sampling rate of 40.6 Hz. For each recording, the photometry signal F was converted to dF/F by $dF/F(t) = (F(t) - \text{median}(F))/\text{median}(F)$.

Rabies virus (RV) tracing (Mice)

We unilaterally injected 0.4 μ l of a mixture of AAV (1:3, AAV-*EF1a-flex-TVA-mCherry*; AAV-CAG-*flex-RG*) into the SubLDT of *Crhbp-Cre* mice. At 3 weeks after AAV injection, we unilaterally injected 1.0 μ l of RV-SAD Δ G-GFP-ERT2-*Cre-ERT2* (EnvA) into the Gi of the same mice. 1 week after RV injection, the mice were killed and their brains were analyzed by histology. We confirmed the specificity of this tracing by the same experiments in WT mice.

Optogenetics (Mice)

For optogenetic experiments, mice were allowed to recover for at least 2 weeks after surgery. The mice were then moved to a sleep recording chamber and connected to an amplifier for EEG/EMG recording. For acclimatization, the mice were connected to an optic patch code (Doric Lenses, Quebec, Canada) for at least 3 days. To stimulate SwiChR++-YFP, blue light (BLM462TA; Shanghai Laser & Optics Century, Shanghai, China: 462 nm, 2–25 mW) was delivered for 4 h during ZT6–ZT10. In each mouse, EEG/EMG was recorded at least twice with/without blue light stimulation, respectively, and the mean value for each condition was calculated.

REMS deprivation with flowerpot (Mice)

The flowerpot procedure was performed following a previously reported protocol⁷⁴ (Figure S2) or⁷⁵ (Figure S4) with slight modification. For Figure S2, mice were subjected to flowerpot for 72 hours with continuous EEG/EMG recording in all groups. For Figure S4, the mice were subjected to flowerpot for 96 h. EEG/EMG recordings were conducted only during the recovery period of the REMS-rebound group, and not in the REMS-deprivation group to protect the amplifier from the water. In both Figures S2 and S4, the mice

were allowed to recover for 30 min in their home cage every day, during which the cage was cleaned. In the REMS-rebound group, the mice were anesthetized and perfused 150 min after detection of the first REMS episode.

Histology (Mice)

Deeply anesthetized mice were killed by injecting a lethal dose of anesthetic and transcardially perfused with saline followed by 10% neutral-buffered formalin. After extraction, the brains were postfixed in the same fixative over at least 1 night and subsequently placed in 30% sucrose (w/v) in phosphate-buffered saline (PBS) until they sank. The brains were rapidly frozen and sectioned at 20–50 μ m using a sliding microtome (REM-710, MC-802A; Yamato Kohki, Asaka, Japan).

Immunohistochemistry (IHC) (Mice)

Unless otherwise stated, the sections were washed 3x 5 min in TBST (1xTBS, Tris-buffered saline (pH 7.5) + 0.1% Tween 20), 3 min 0.3% H₂O₂/1xTBST, and again 3x 5 min in TBST. At 30 min after incubation in blocking buffer (1xTBS + 0.5% Blocking Reagent (FP1020; Perkin Elmer, Waltham, MA, USA)), the sections were incubated overnight with a primary antibody for red fluorescent protein (RFP) (1/500 rabbit anti-RFP (PM006; Medical & Biological Laboratories, Nagoya, Japan), goat anti-tdTomato (1/2000 AB8181-200; SICGEN, Cantanhede, Portugal)), GFP (1/500 rat anti-GFP (GF090R; Nacalai Tesque, Kyoto, Japan)), Cre (1/500 rabbit anti-Cre (69050-3; Novagen, Burlington, MA, USA)), NeuN (1/1000 mouse anti-NeuN (MAB377; Millipore, Burlington, MA, USA)), TH (1/100 mouse anti-TH (T2928; Sigma-Aldrich, St. Louis, MO, USA)), CRHBP (1:1000 rabbit anti-CRHBP (human); aa98-136, produced in this study (see below)), CRHBP (1:1000 goat anti-CRHBP (human); aa25-322, AF2796, R&D Systems, Minneapolis, MN, USA) at 4°C or for choline acetyltransferase (ChAT) (1/100 goat anti-ChAT (AB144P; Millipore, Burlington, MA, USA)) or c-fos (1/2000: rabbit anti-c-fos (ABE457; Millipore, Burlington, MA, USA) at room temperature (RT). After 3x 5 min wash in TBST, the sections were incubated with a secondary antibody (1/500 donkey anti-rabbit HRP (ab7083; Abcam, Cambridge, UK), 1/500: donkey anti-rat HRP (ab102182; Abcam, Cambridge, UK), or CF-conjugated antibodies (1/500 Biotium, Fremont, CA, USA)) and 1 μ g/ml 4',6-diamidino-2-phenylindole (DAPI) for 2 h. After 4x 5 min wash in TBST, the sections were incubated with tyramide-conjugated fluorescein, Cy3, or Cy5 (Perkin Elmer, Waltham, MA, USA) for 30 min. Finally, after washing 4x 5 min in TBST, the sections were mounted on slideglass using mounting medium (PermaFluor TA-006-FM; Eprelia, Kalamazoo, MI, USA).

For co-staining with c-fos and Rpl10a-GFP (Figure S2), we applied 3,3'-diaminobenzidine tetrahydrochloride (DAB)-Nickel (Ni) chromogenic detection. After incubation in 0.3% H₂O₂/1xTBST and wash in TBST, the sections were incubated in primary antibody for c-fos (1/3000 rabbit anti-c-fos (ABE457; Millipore, Burlington, MA, USA)) or GFP (1/20000 rat anti-GFP (GF090R; Nacalai Tesque, Kyoto, Japan)) for overnight at RT. After washing in TBST, the sections were incubated in biotinylated secondary antibodies for 1 h (1/500 711-065-152 (for rabbit) or 712-065-153 (for rat); Jackson ImmunoResearch, West Grove, PA, USA), followed by incubation in horseradish peroxidase (HRP)-conjugated avidin-biotin complex solution for 1 h (1/1000 PK-6100; VECTOR Laboratories, Burlingame, CA, USA). We visualized c-fos signals with the black products produced by DAB plus nickel (146-01012; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) and Rpl10a-GFP signals with the brown products produced by DAB (349-00903; Dojindo, Kamimashiki, Japan). After mounting on slideglass and incubation in EtOH (dehydration) and xylene (clearing), sections were cover-slipped with Entellan new (1.07961.0100; Merck, Burlington, MA, USA).

Images were captured with a laser confocal microscope (LSM700, LSM800; Carl Zeiss microscopy, Jena, Germany), or slide scanner (NanoZoomer XR; Hamamatsu Photonics, Hamamatsu, Japan, or AxioScan.Z1; Carl Zeiss microscopy, Jena, Germany).

In situ hybridization (ISH)/IHC double staining (Mice)

For co-localization analysis of hM3Dq and verification of DTA-mediated ablation of *Crhbp* neurons, we applied *in situ* hybridization using antisense digoxigenin (DIG)-labeled RNA probes. Template DNA sequences of *Vglut2*, *Vgat*, *Crhbp* were produced mimicking those designed in the Allen Brain Atlas (<http://mouse.brain-map.org/>) (*Vglut2*; RP_050921_01_E03, *Vgat*; RP_07124_02_A05, *Crhbp*; RP_070129_01_F08).⁷⁶

Sections were incubated in 10% neutral-buffered formalin for 15 min and washed 3x 5 min with PBST (1xPBS + 0.1% Tween 20). After 10 min incubation in 10 μ g/ml proteinase K (03115887001; Roche, Basel, Switzerland) at 37°C, the sections were fixed in 10% neutral-buffered formalin for 20 min and washed 3x 5 min with PBST. DIG-labeled RNA probes (1 μ g/ml) in 50% formamide/5x standard saline citrate (SSC)/0.1% Tween 20 were added to the sections and incubated at 60°C overnight. The sections were washed with 50% formamide/2x SSC/0.1% Tween 20 at 60°C for 60 min, 2x SSC/0.1% Tween 20 at 60°C for 15 min, and 0.2x SSC/0.1% Tween 20 at 60°C for 60 min. The sections were then washed 3x 5 min with TBST, and incubated overnight with anti-alkaline phosphatase labeled DIG antibody (1/1,000 anti-DIG-alkaline phosphatase (11093274910; Roche, Basel, Switzerland)) at 4°C 1 overnight and washed 3x 5 min in TBST. For detection of *Vglut2* and *Vgat*, after 3x wash in 0.1M Tris-HCl (pH8.2)/0.1% Tween 20, sections were incubated in the HNPP/Fast Red solution (11758888001; Roche, Basel, Switzerland) for 0.5–2 h and washed again in TBST 3x 5 min. For detection of *Crhbp*, after 3x wash in 0.1M Tris-HCl (pH9.0)/0.1M NaCl/0.1% Tween 20, sections were incubated in the BCIP/NBT solution (11681451001; Roche, Basel, Switzerland) for 0.5–3 h and washed in TBST 3x 5 min. Subsequently, IHC for mCherry or GFP was conducted. The sections were incubated overnight with primary antibodies for RFP (1/500: goat anti-tdTomato (AB8181-200; SICGEN, Cantanhede, Portugal)) or denatured GFP (1/100: rabbit anti-denatured GFP (SC8344; Santa Cruz Biotechnology, Dallas, TX, USA)) at RT, followed by secondary antibody staining using CF-conjugated antibodies (Biotium, Fremont, CA, USA) and DAPI.

Images were captured with a laser confocal microscope (LSM700, LSM800; Carl Zeiss microscopy, Jena, Germany) or a slide scanner (NanoZoomer XR; Hamamatsu Photonics, Hamamatsu, Japan, or AxioScan.Z1; Carl Zeiss microscopy, Jena, Germany).

For co-localization analysis of *Crhbp-Cre*; *LSL-Rpl10a-GFP* in the SubLDT (Figures S1C and S1D), *Crhbp*^{SubLDT→BF} neurons, *CamkIIa*^{DPGi,Pr}, *Crh*^{DPGi,Pr}, and *Nos1*^{DPGi,Pr} neurons (Figures 4J and S5A–S5D), we used an RNAscope Multiplex Fluorescent Reagent Kit v2 (323110; Advanced Cell Diagnosis, Newark, CA, USA) following the manufacturer's protocol. The probes used were *Vglut2*; 456751-C2, *Vgat*; 319191-C4, *Crhbp*; 498971-C1, *Crh*; 316091-C1, and *Nos1*; 437651-C1. GFP or RFP was visualized by IHC using an antibody for denatured GFP (1/100 rabbit anti-denatured GFP (SC8344; Santa Cruz Biotechnology, Dallas, TX, USA)) or RFP (1/500: goat anti-tdTomato (AB8181-200; SICGEN, Cantanhede, Portugal) followed by a CF-conjugate antibody (BIOTIUM, Fremont, CA, USA) and DAPI. Images were captured with a slide scanner (AxioScan.Z1; Carl Zeiss microscopy, Jena, Germany).

Generation of human CRHBP antibody

In the present study, we produced CRHBP antibody in rabbit. For expression of glutathione S-transferase fusion proteins, we subcloned cDNA fragments encoding the corresponding 98–136 amino acid residues of human CRHBP (GenBank #CR541666.1) into the BamHI/EcoRI site of pGEX4T-2 plasmid (GE Healthcare, Buckinghamshire, UK). Immunization and affinity purification were performed as reported previously.⁷⁷

Cell culture and overexpression or siRNA for *Crhbp* for Western blotting

293A human kidney cells (R70507; Thermo Fisher Scientific, Waltham, MA, USA) were cultured in Dulbecco's modified Eagle's medium (043-30085; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) supplemented with 10% fetal bovine serum (35-079-CV; CORNING, Corning, NY, USA), 100 units/ml penicillin and 100 µg/ml streptomycin (09367-34; Nacalai Tesque, Kyoto, Japan). Cells were cultured at 37°C in a 5% CO₂ incubator. Human CRHBP expression vector (#FHC22203, Promega, Madison, WI, USA) and the siRNA targeted to human CRHBP (siCRHBP, sc-39397, Santa Cruz Biotechnology, Dallas, TX, USA) were used in this study. CRHBP expression vector was transfected into 293A cells using FuGENE6 (Promega, Madison, WI, USA). siCRHBP was transfected using LipofectamineTM RNAiMAX Transfection Reagent (13778075, Thermo Fisher Scientific, Waltham, MA, USA) according to the manufacturer's instructions. Cells were collected 48 h after transfection.

Western blotting

Post-mortem human brain samples derived from the cortex and cultured cells were lysed in the lysis buffer (25 mM HEPES (pH7.5), 150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% sodium dodecyl sulfate, 0.1 M NaF, 10mM sodium pyrophosphate, 10% glycerol, 1 mM dithiothreitol, 0.15 mM spermine, 2 mM spermidine, 10 µg/ml leupeptin, 12.5 µg/ml ALLN, 2µg/ml aprotinin, and 5µg/ml pepstatin A). The total cell lysates were separated by SDS polyacrylamide gel electrophoresis and transferred to polyvinylidene difluoride membranes. The membranes were incubated with the antibodies for CRHBP and GAPDH (014-25524; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) as a control. Bound antibodies were visualized with horseradish peroxidase-conjugated IgG, and ImmunoStar LD (290-69904; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) or Chemi-Lumi One L (07880-70; Nakalai Tesque, Kyoto Japan), and detected by ChemiDoc XRS+ (BioRad, Hercules, CA, USA).

Immunocytochemistry

293T human kidney cells (RCB2202; Riken BRC, Tsukuba, Ibaraki, Japan) were cultured in Dulbecco's modified Eagle's medium (043-30085; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan) supplemented with 10% (v/v) fetal bovine serum (35-010-CV; CORNING, Corning, NY, USA), and 1% (v/v) penicillin/streptomycin (168-23191; FUJIFILM Wako Pure Chemical Corporation, Osaka, Japan). Cells were cultured at 37°C in a 5% CO₂ incubator. The full-length human CRHBP expression vector (#FHC22203, Promega, Madison, WI, USA) was used. Plasmids expressing CRHBP Δaa98-136 or CRHBP Δaa25-322 were generated by replacing full-length CRHBP with PCR fragments carrying the above truncated forms of CRHBP using PvuI and PmeI sites. 293T cells were transfected with either of the CRHBP-expressing vectors using PEI-Max (24765-1, Polysciences, Warrington, PA, USA). 48–72 h after transfection, cells were subjected to immunocytochemistry.

Cells were washed with PBS, fixed with 4% paraformaldehyde/PBS (w/v) for 20min, washed with PBS, permeabilized with 1% TritonX-100/PBS (v/v) for 20min, and washed again with PBS. 30 min after incubation in blocking buffer (1xTBS + 0.5% Blocking Reagent (FP1020; Perkin Elmer, Waltham, MA, USA), the cells were incubated for 2 h with the primary antibodies at room temperature. After washing with PBS, the cells were treated with CF-conjugated antibodies (1/1000 BIOTIUM, Fremont, CA, USA) and 1 µg/ml DAPI for 2 h. After washing in PBS, the cells were mounted on slideglass using a mounting medium (PermaFluor TA-006-FM; Eprelia, Kalamazoo, MI, USA). Images were captured with a laser confocal microscope (LSM800; Carl Zeiss microscopy, Jena, Germany). Images were analyzed by ImageJ.⁷⁸ Binarized DAPI images were used to determine the region of interest (ROI), which were then applied to the other channel images. For quantification, the mean value within each ROI was calculated.

Immunohistochemistry (IHC) (Human)

In each case, the left hemisphere of the brain including the brainstem was fixed in 10% buffered formalin for pathologic assessment. The rest of the brain tissue was frozen for molecular and biochemical analyses. After macroscopic observation of formalin-fixed brain

tissues, the upper portion of the pons were cut at 5-mm intervals and the samples were embedded in paraffin. Serial sections of 6- μ m thickness were stained with luxol fast blue/cresyl violet (also known as Klüver-Barrera staining) (Figure S6). For immunohistochemistry, deparaffinized sections were incubated for 30 min with 0.3% hydrogen peroxide to quench endogenous peroxidase activity, then washed with PBS. The primary antibodies used were rabbit polyclonal antibodies against DBH (1:500; ab209487, abcam, Cambridge, UK), ChAT (1:500; AB143, Millipore, Burlington, MA, USA), CRHBP (1:100; aa98-136, produced in this study), and a goat polyclonal antibody against CRHBP (1:50; AF2796, R&D Systems, Minneapolis, MN, USA). Autoclave treatment was performed for 15 min before incubation. Goat anti-rabbit or anti-mouse immunoglobulins conjugated to peroxidase-labeled dextran polymer (Dako Envision+, Dako, Glostrup, Denmark) and horse anti-goat IgG antibody (H+L) (BA-9500, VECTOR Laboratories, Burlingame, CA, USA) were used as secondary antibodies. Reaction products were visualized using DAB (ImmPACT DAB, VECTOR laboratories, Burlingame, CA, USA). Hematoxylin was used to counterstain the cell nuclei.

For double immunohistochemistry, 2 primary antibodies were combined, including a rabbit polyclonal antibody against CRHBP (1:100; aa98-136, produced in this study) and a mouse monoclonal antibody against p- α -syn (1:1000; clone pSyn#64, Fujifilm-Wako Industrial, Osaka, Japan). Autoclave treatment was performed for 15 min before incubation with the primary antibodies. Goat anti-rabbit and anti-mouse immunoglobulins conjugated to peroxidase-labeled dextran polymer (Dako Envision+, Dako, Glostrup, Denmark) were used as secondary antibodies. Reaction products were visualized using ImmPACT DAB and Stayright™ Purple (45906, AAT Bioquest, Sunnyvale, CA, USA).

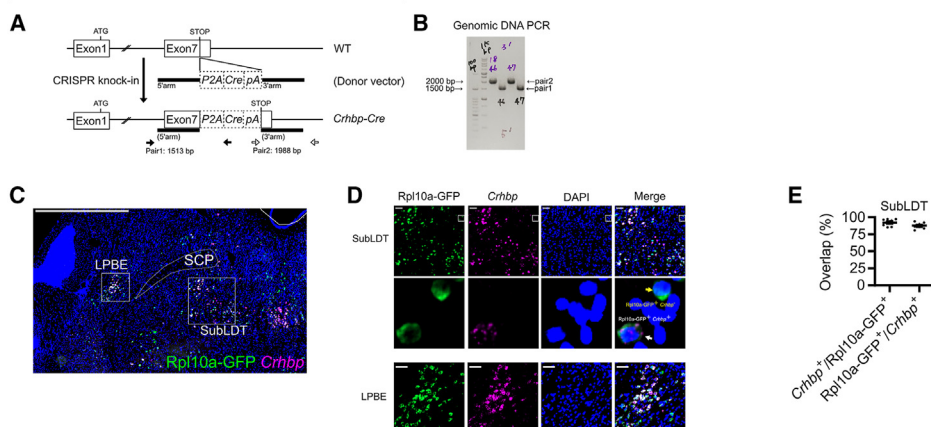
For cell counting, 12 images were obtained using a digital camera connected to a microscope (Eclipse Ni-E, Nikon, Tokyo, Japan) in each case, and ChAT- or CRHBP-IR neurons were counted (see also Figure S6).

QUANTIFICATION AND STATISTICAL ANALYSIS

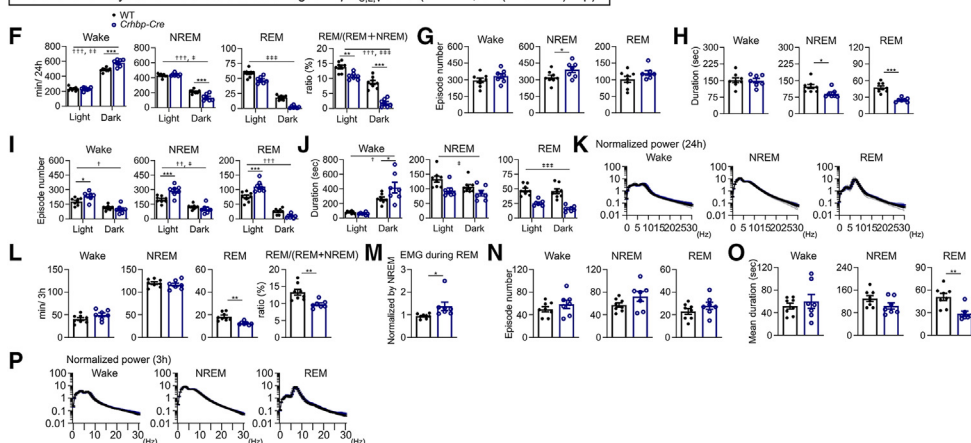
The statistical details of the experiments are provided in the figure legends. The sample size was determined based on previous relevant studies. Graph represents mean $\pm$ SEM. Data were analyzed with Prism (Graph Pad, San Diego, CA, USA). A p-value of less than 0.05 was considered statistically significant. Where applicable, all tests were two-tailed. Bootstrap in Figure 7 was conducted by R (version 4.3.2, R Foundation for Statistical Computing, Vienna, Austria) and RStudio (version 2023.12.0, RStudio: Integrated Development for R. RStudio, PBC, Boston, MA), using a package of bootstrap (version: 2019.6). In each panel, we first mixed all the data. Then, resampling was conducted 2000 times and the bootstrap p-value was calculated based on the sample mean.

Supplemental figures

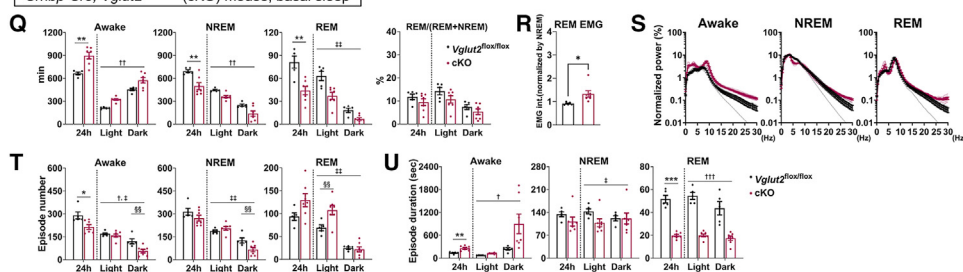
Generation and characterization of *Crhbp-Cre* mouse



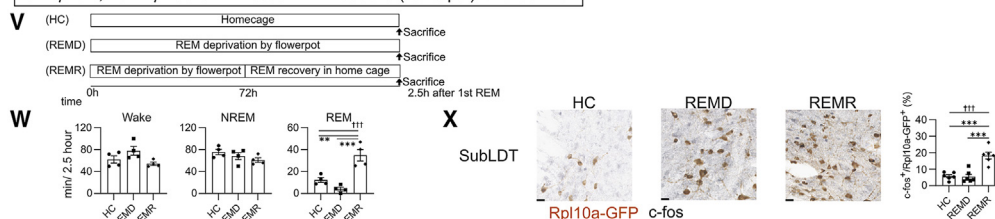
Additional analyses of the effects of ablating *Crhbp*⁺ cells (24h: f-k, 3h (ZT5-ZT7): l-p)



Crhbp-Cre; *Vglut2*^{flax/flax} (cKO) mouse; basal sleep



Crhbp-Cre; LSL-*Rpl10a-GFP* mouse: REM rebound (flowerpot) + c-fos IHC



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Figure S1. Generation of *Crhbp*-Cre mice, additional analyses related to the ablation of *Crhbp*⁺_{S,L,V} neurons, and c-fos immunohistochemistry of *Crhbp*⁺_{SubLDT} neurons following REMS rebound induced by the flowerpot method, related to Figure 1

(A) *P2A-Cre-pA* was inserted immediately before the stop codon of *Crhbp* to generate *Crhbp*-Cre mice. Arrows indicate primer pairs used for polymerase chain reaction (PCR) to confirm insertion of the transgene.

(B) Insertion of the transgene was confirmed by genomic DNA PCR.

(C–E) Co-expression analysis of GFP (*Crhbp*-Cre; *LSL-Rpl10a-GFP*) with *Crhbp* (ISH) in SubLDT (D and E) and external lateral parabrachial nucleus (LPBE) (D). Data represent mean ± SEM. *n* = 8 sections, obtained from 4 mice. Scale bar: 1 mm in (C) and 0.05 mm in (D).

(F–P) Sleep architecture (F–J: 24 h, L–O: 3 h [ZT5–ZT7]) and EEG power spectra (K: 24 h, P: 3 h [ZT5–ZT7]) of WT or *Crhbp*-Cre mice infected with AAV1-*mCherry-flex-DTA* (related to Figures 1I–1K). Data represent mean ± SEM. *n* = 8 mice in WT group, *n* = 7 mice in *Crhbp*-Cre group. (F, I, and J) *,†,‡*p* < 0.05; **, ††,‡‡*p* < 0.01; ***,†††,‡‡‡*p* < 0.001 (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test). (G, H, and L–O) **p* < 0.05; ****p* < 0.001 (unpaired *t* test).

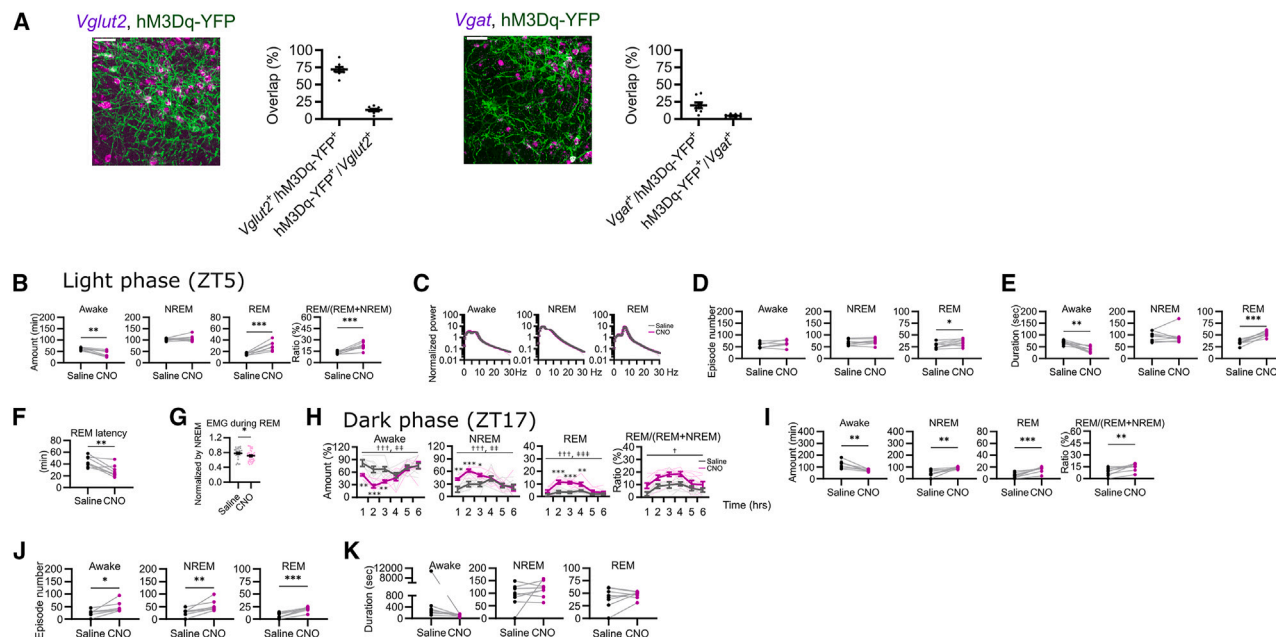
(Q–U) Sleep architecture (Q, T, and U), EMG integral during REMS (R), and EEG power spectra (S) of *Crhbp*-Cre; *Vglut2*^{flax/flax} (cKO) and control *Vglut2*^{flax/flax} mice. Data represent mean ± SEM. *n* = 5 mice in the control group, *n* = 7 mice in the cKO group. (Q, T, and U) †,‡*p* < 0.05; §§,††,‡‡*p* < 0.01; †††*p* < 0.001 (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and § indicates significance in Bonferroni's test). (R) **p* < 0.05; ***p* < 0.01; ****p* < 0.001 (unpaired *t* test).

(V) Schematic diagram of the experimental conditions for flowerpot.

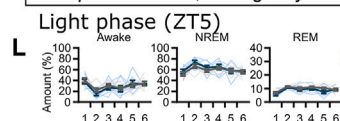
(W) Time spent in each vigilance state during 2.5 h before the mice were killed. Data represent mean ± SEM, *n* = 4 mice/group. ***p* < 0.01; ***,†††*p* < 0.001 (one-way ANOVA followed by Tukey's test; † indicate significant difference in one-way ANOVA, and * indicates significance in Tukey's test).

(X) c-fos (black signals developed by 3,3'-diaminobenzidine tetrahydrochloride [DAB] plus nickel) and *Rpl10a*-GFP (brown signals developed by DAB) signals were detected in SubLDT. Data represent mean ± SEM, *n* = 6 mice/group. Scale bar: 20 μm. ***,†††*p* < 0.001 (one-way ANOVA followed by Tukey's test; † indicate significant difference in one-way ANOVA, and * indicates significance in Tukey's test).

Crhbp-Cre mouse; AAVrg-hSyn-DO-mCherry-DIO-flp (Gi), AAV8-hSyn-fDIO-hM3Dq-YFP (SubLDT)



Crhbp-Cre mouse; AAVrg-hSyn-DO-mCherry-DIO-flp (Gi), AAV8-hSyn-fDIO-jGCaMP7s (SubLDT)



Crhbp-Cre mouse; AAVrg-DO-mCherry-DIO-flippase (BF) / AAV8-fDIO-hM3Dq-YFP (SubLDT)

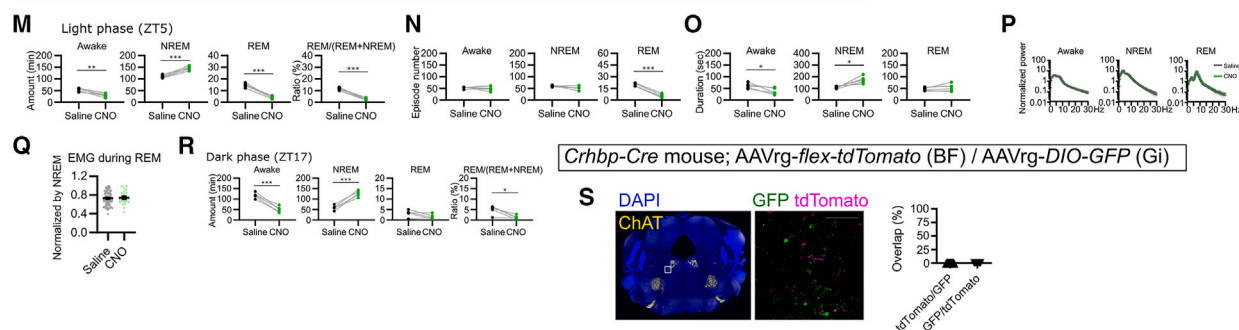


Figure S2. Additional analyses on the effect of manipulation of *Crhbp*⁺_{SubLDT→Gi} neurons and *Crhbp*⁺_{SubLDT→BF} neurons, related to Figures 3 and 4

(A) Co-expression analysis of hM3Dq-YFP with *Vglut2* (ISH) or *Vgat* (ISH), respectively. Data represent mean $\pm$ SEM. $n = 8$ sections, obtained from 4 mice. (Scale bar: 0.05 mm; green, hM3Dq-YFP; magenta, *Vglut2* [ISH] or *Vgat* [ISH]).

(B–K) Sleep architecture (B, D, E, and H–K), EEG power spectra (C), and mean EMG integral values of each REMS episode normalized by those during NREMS episodes (G) during the 3 h following saline or CNO (1.5 mg/kg) administration during the light phase (ZT5; B–G) or dark phase (ZT17; H–K) (related to Figures 3A–3D). REMS latency (F) represents the latency to enter the first REMS episode following drug administration. Data represent mean $\pm$ SEM (C, G, and H). $n = 8$ mice (A–F and H–K). $n = 32$ (saline) or 46 (CNO) episodes from 8 mice (G).

(L) Time spent in each vigilance state after saline or CNO (1.5 mg/kg) administration during the light phase (ZT5) in *Crhbp-Cre* mice expressing jGCaMP7s. Horizontal axes indicate time after drug administration. Data represent mean $\pm$ SEM. $n = 8$ mice.

(B, D–G, and I–K) $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$ (paired Student's t test). (H and L) $^*p < 0.05$; $^{**}, \ddagger, \ddagger\ddagger, \ddagger\ddagger\ddagger p < 0.01$; $^{***}, \ddagger, \ddagger\ddagger, \ddagger\ddagger\ddagger p < 0.001$ (two-way repeated-measures ANOVA followed by Bonferroni's test; $\ddagger$ and $\ddagger$ indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and $*$ indicates significance in Bonferroni's test).

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(M–R) Sleep architecture (M–O and R), EEG power spectra (P), and mean EMG integral values of each REMS episode normalized by those during NREMS episodes (Q) during the 3 h following saline or CNO (1.5 mg/kg) administration at the light phase (ZT5; M–O) or dark phase (ZT17; R). Data represent mean $\pm$ SEM (P and Q). $n = 5$ mice/group (M–P and R). $n = 85$ (saline) or 24 (CNO) episodes obtained from 5 mice (Q). * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$ (paired Student's t test).

(S) 2 different retrograde tracers were each microinjected into BF (AAV.rg-CAG-*flex-tdTomato*) and Gi (AAV.rg-*hSyn-DIO-GFP*) of *Crtbpc-Cre* mice. The graph represents the quantification of double-positive cells in SubLDT. $n = 6$ sections obtained from 3 mice. Data represent mean $\pm$ SEM. Scale bar: 0.1 mm.

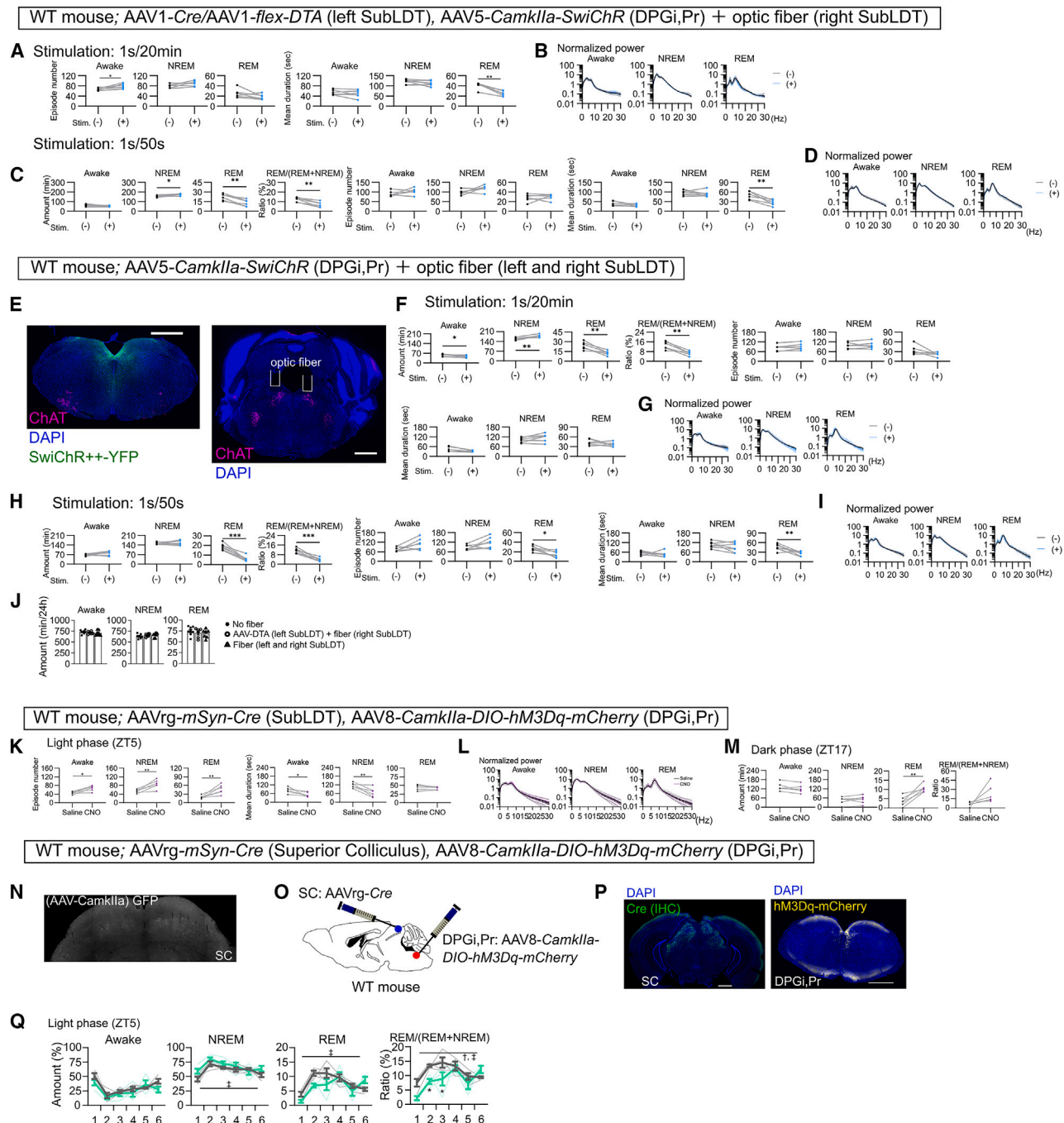


Figure S3. Additional analyses on the effect of manipulation of *CamkIIa*⁺ DPGi,Pr→SubLDT neurons, related to Figure 5

(A–D) Sleep architecture (A: stimulation 1 s/20 min; C: stimulation 1 s/50 s) and EEG power spectra (B: stimulation 1 s/20 min; D: stimulation 1 s/50 s) with or without unilateral optogenetic inhibition of *CamkIIa*⁺ DPGi,Pr→SubLDT neuron axon terminals for 4 h in the light phase (ZT6–ZT10) in mice with unilateral ablation of SubLDT neurons (related to Figures 5F–5I). *n* = 6 mice/group. **p* < 0.05; ***p* < 0.01 (paired Student's *t* test).

(E) AAV-encoding SwiChR⁺-YFP was microinjected into the DPGi/Pr of WT mice. Optic fibers were bilaterally inserted above SubLDT. (Scale bar: 1 mm; green, SwiChR⁺-YFP; magenta, ChAT; blue, DAPI).

(F–I) Sleep architecture (F: stimulation 1 s/20 min; H: stimulation 1 s/50 s) and EEG power spectra (G: stimulation 1 s/20 min; I: stimulation 1 s/50 s) with or without bilateral optogenetic inhibition of *CamkIIa*⁺ DPGi,Pr→SubLDT neuron axon terminals for 4 h in the light phase (ZT6–ZT10). *n* = 6 trials from 3 mice/group (F and G). *n* = 7 mice/group (H and I). **p* < 0.05; ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test).

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(J) Basal amount of sleep or wake during 24 h in mice that underwent AAV injection in DPGi/Pr plus unilateral lesion/fiber insertion in SubLDT (related to A, D, and [Figures 5F–5I](#)) and mice that underwent AAV injection in DPGi/Pr plus bilateral fiber insertion above SubLDT (related to F–I) compared with mice that solely underwent AAV injection in DPGi/Pr. $n = 8$ (AAV only), 6 (AAV + lesion + fiber), 7 (AAV + bilateral fiber) mice. Data represent mean $\pm$ SEM (one-way ANOVA).

(K–M) Sleep architecture (K and M) and EEG power spectra (L) during the 3 h following saline or CNO (1.5 mg/kg) administration at ZT5 (K) or ZT17 (M) in mice expressing hM3Dq-mCherry in *CamkIIa*^{DPGi,Pr→SubLDT} neurons (related to [Figures 5J–5M](#)). $n = 6$ mice. * $p < 0.05$; ** $p < 0.01$ (paired Student's *t* test).

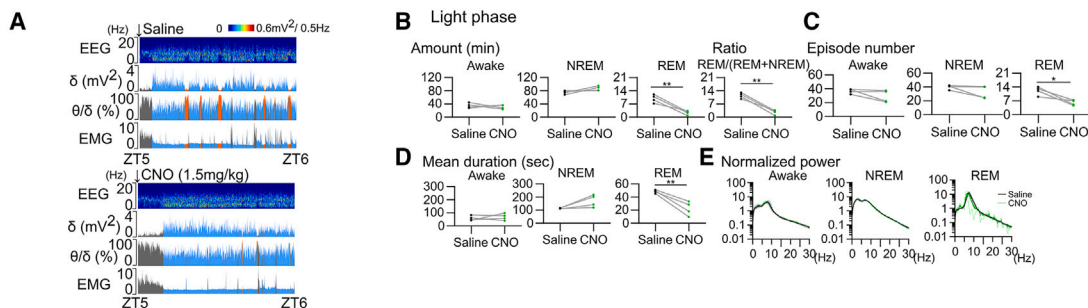
(N) GFP-labeled axons were detected in superior colliculus (SC) when AAV-encoding GFP was microinjected into DPGi/Pr.

(O) Retrograde AAV-encoding Cre was microinjected into SC, and AAV-encoding Cre-dependent hM3Dq-mCherry was microinjected into DPGi/Pr of WT mice to express hM3Dq in *CamkIIa*^{DPGi,Pr→SC} neurons.

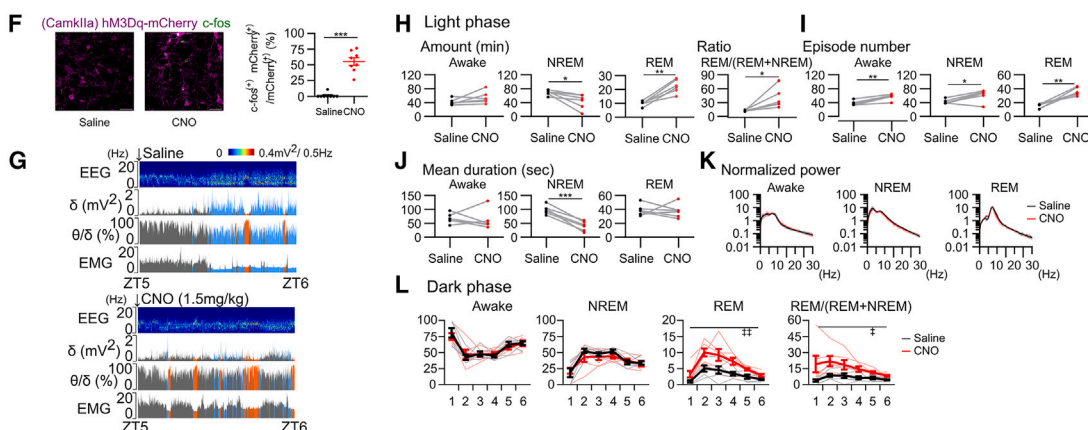
(P) hM3Dq-mCherry and Cre were detected in DPGi/Pr and SC, respectively. (Scale bar: 1 mm; green, Cre; yellow, hM3Dq-mCherry; blue, DAPI).

(Q) Time spent in each vigilance state after chemogenetic activation of *CamkIIa*^{DPGi,Pr→SC} neurons. Drug administration (saline or CNO [1.5 mg/kg]) was performed in the light phase (ZT5). Horizontal axes indicate time after drug administration. Data represent mean $\pm$ SEM. $n = 4$ mice/group. †, ‡, * $p < 0.05$ (two-way repeated-measures ANOVA followed by Bonferroni's test; † and ‡ indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and * indicates significance in Bonferroni's test).

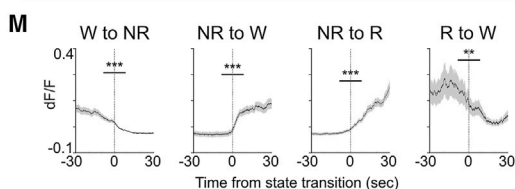
WT mouse; AAV8-CamkIIa-hM4Di-mCherry (DPGi,Pr)



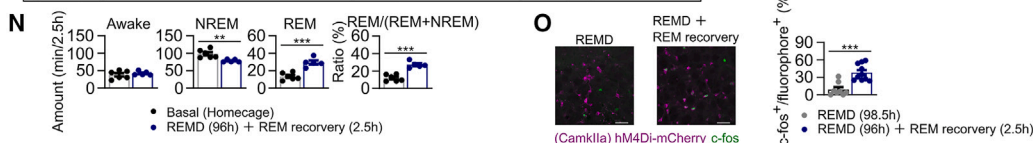
WT mouse; AAV8-CamkIIa-hM3Dq-mCherry (DPGi,Pr)



Additional analysis of fiber photometry [WT mouse + AAV-CamkIIa-GCaMP6s (DPGi,Pr)]

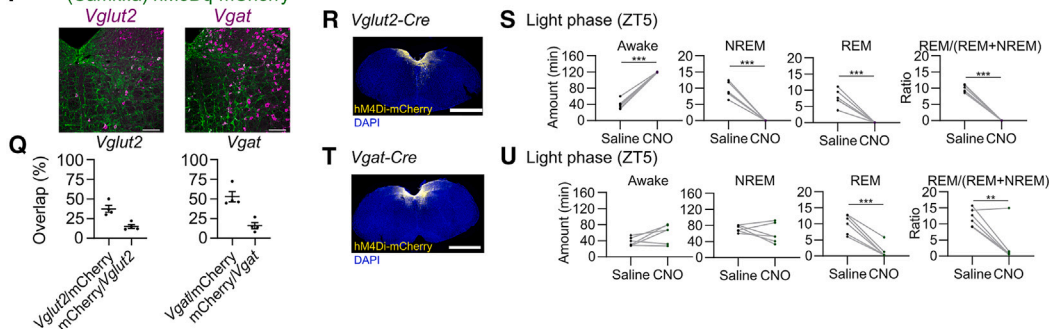


[DPGi,Pr] WT mouse + AAV-CamkIIa-fluorophore; REM rebound (flowerpot) + c-fos IHC



(CamkIIa) hM3Dq-mCherry

Vglut2-Cre or Vgat-Cre mouse; AAV8-CamkIIa-DIO-hM4Di-mCherry (DPGi,Pr)



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Figure S4. Additional analyses on the effect of manipulation of *Camklla*^{DPGi,Pr} neurons, fiber photometry, and c-fos immunohistochemistry, related to Figure 5

(A–L) (A–E) Results of mice expressing hM4Di-mCherry; (F–L) results of mice expressing hM3Dq-mCherry, respectively (related to Figures 5M–5O).
(A and G) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral following administration of saline or CNO (1.5 mg/kg) in mice expressing hM4Di-mCherry (A) or hM3Dq-mCherry (G) in *Camklla*^{DPGi,Pr} neurons (related to Figures 5M–5O).
(B–E and H–K) Sleep architecture (B–D and H–J) and EEG power spectra (E and K) during the 2 h following saline or CNO (1.5 mg/kg) administration at ZT5 in mice expressing hM4Di-mCherry (B–E) or hM3Dq-mCherry (H–K) in *Camklla*^{DPGi,Pr} neurons (related to Figures 5M–5O). (B–E) *n* = 4 mice; (H–K) *n* = 6 mice/group. **p* < 0.05; ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test). (E, L, and Q) Data represent mean ± SEM.
(F) c-fos immunohistochemistry of mice expressing hM3Dq-mCherry in *Camklla*^{DPGi,Pr} neurons that were killed following saline or CNO (4.5 mg/kg) administration. Data represent mean ± SEM. *n* = 8 sections/group obtained from 4 mice/group.
(L) Time spent in each vigilance state following saline or CNO (1.5 mg/kg) administration during the dark phase (ZT17). Data represent mean ± SEM. *n* = 6 mice. ‡*p* < 0.05; ††*p* < 0.01 (two-way repeated-measures ANOVA followed by Bonferroni's test); ‡ indicate significant main effect of intervention, in two-way repeated-measures ANOVA.
(M) Photometry signals aligned to vigilance state transitions in mice expressing GCaMP6s in *Camklla*^{DPGi,Pr} neurons (related to Figures 5P–5S). Data represent mean ± SEM. Wake to NREMS, *n* = 41 episodes obtained from 8 mice; NREMS to wake, *n* = 28 episodes obtained from 8 mice; NREMS to REMS, *n* = 64 episodes obtained from 8 mice; REMS to wake, *n* = 12 episodes obtained from 8 mice; ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test).
(N and O) Co-expression analysis of hM3Dq-mCherry with *Vglut2* (ISH) or *Vgat* (ISH) in WT mice in which *Camklla*^{DPGi,Pr} neurons were labeled by microinjecting AAV encoding either GFP or hM4Di-mCherry into the DPGi/Pr of WT mice. Time spent in each vigilance state in the basal (*n* = 6 mice) and REMS recovery group (*n* = 5 mice) (N), representative image of c-fos immunohistochemistry (O, left) (scale bar: 0.05 mm; green, c-fos; magenta, mCherry), and comparison of the ratio of c-fos/mCherry double-positive cells to mCherry positive cells (O, right). *n* = 8 sections obtained from 4 mice in REMS-deprived group, *n* = 10 sections obtained from 5 mice in REMS-rebound group. ***p* < 0.01; ****p* < 0.001 (unpaired Student's *t* test).
(P and Q) Co-expression analysis of hM3Dq-mCherry with *Vglut2* (ISH) or *Vgat* (ISH) in WT mice in which *Camklla*^{DPGi,Pr} neurons were labeled by microinjecting AAV-encoding hM3Dq-mCherry into DPGi/Pr. Data represent mean ± SEM. *n* = 4 sections obtained from 4 mice. (Scale bar: 0.1 mm; green, hM3Dq-mCherry; magenta, *Vglut2* [ISH] or *Vgat* [ISH].)
(R–U) Chemogenetic inhibition of glutamatergic (R and S) or GABAergic (T and U) *Camklla*^{DPGi,Pr} neurons by microinjecting AAV-encoding Cre-dependent hM4Di-mCherry (AAV.8-*Camklla-DIO-hM4Di-mCherry*) into DPGi/Pr of *Vglut2-Cre* (R) or *Vgat-Cre* (T) mice, respectively. (Scale bar: 1 mm; blue, DAPI; yellow, hM4Di-mCherry.) Time spent in each vigilance state during the 2 h following saline or CNO (1.5 mg/kg) administration at the light phase (ZT5) in *Vglut2-Cre* (S) or *Vgat-Cre* (U) mice. *n* = 6 mice/group. ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test).

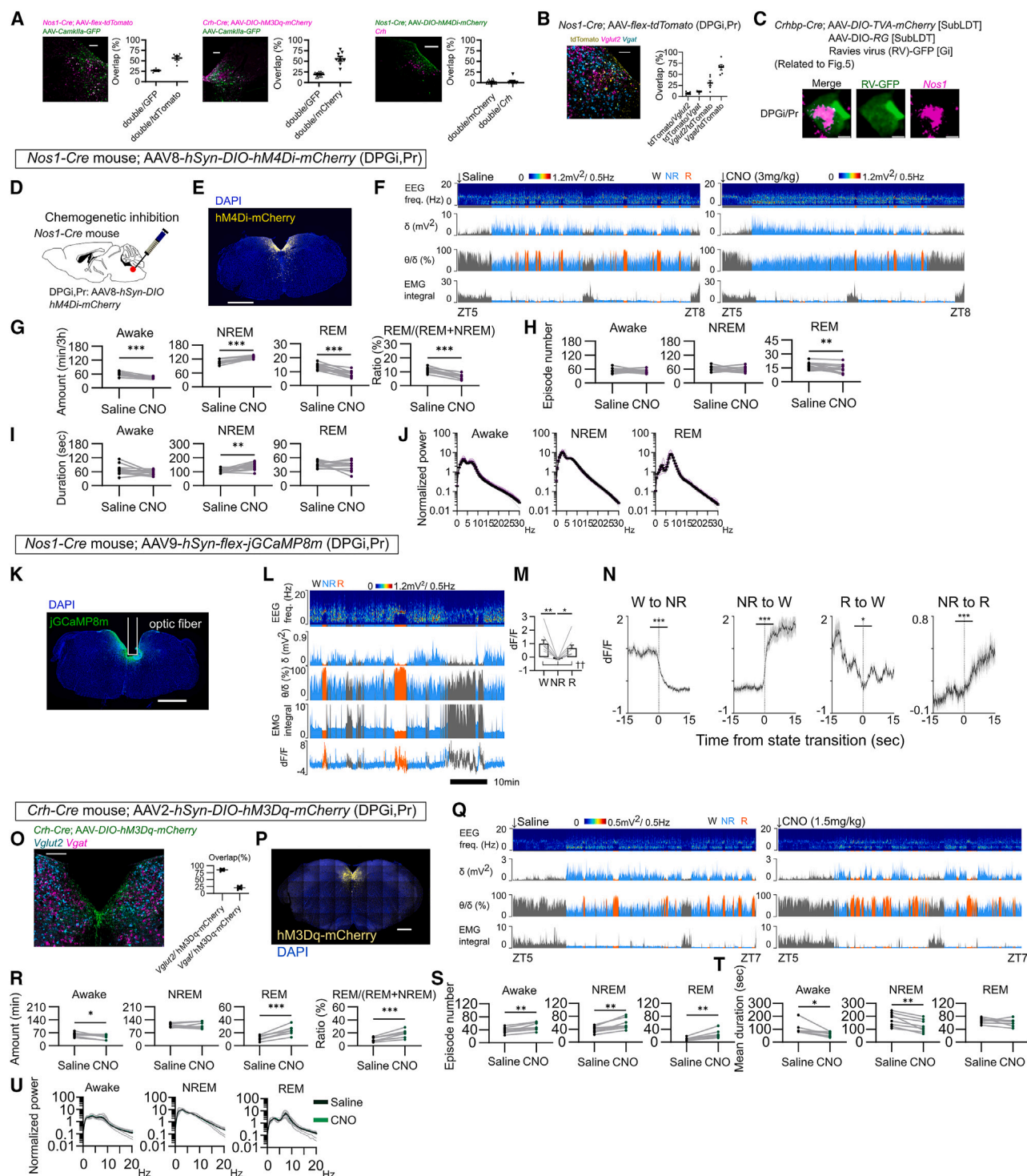


Figure S5. Additional analyses of *Nos1*⁺ DPGi,Pr neurons and *Crh*⁺ DPGi,Pr neurons, related to Figure 6

(A) Analyses of overlap among *CamkIIa*⁺ DPGi,Pr neurons, *Nos1*⁺ DPGi,Pr neurons, and *Crh*⁺ DPGi,Pr neurons. Data represent mean $\pm$ SEM. (Left, scale bar: 200 μ m) magenta, *Nos1-Cre* + AAV-flex-tdTomato; green, AAV-CamkIIa-GFP. *n* = 6 sections obtained from 3 mice. (Middle, scale bar: 100 μ m) magenta, *Crh-Cre* + AAV-DIO-hM3Dq-mCherry; green, AAV-CamkIIa-GFP. *n* = 8 sections obtained from 4 mice. (Right, scale bar: 200 μ m) magenta, *Crh* (ISH); green, *Nos1-Cre* + AAV-DIO-hM4Di-mCherry. *n* = 8 sections obtained from 4 mice.

(B) Co-expression analysis of tdTomato (which labels *Nos1*⁺ DPGi,Pr neurons) with *Vglut2* (ISH) or *Vgat* (ISH). Data represent mean $\pm$ SEM. *n* = 6 sections, obtained from 3 mice. (Scale bar: 0.1 mm; yellow, tdTomato; magenta, *Vglut2* [ISH]; cyan, *Vgat* [ISH]).

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(C) Co-expression analysis of RV-GFP (which labels DPGi/Pr neurons that directly project to *Crhbp*⁺_{SubLDT→Gi} neurons) and *Nos1* (ISH). (Scale bar: 10 μ m; green, RV-GFP; magenta, *Nos1* [ISH]). (Related to [Figures 5A–5D](#)).

(D and E) Chemogenetic inhibition of *Nos1*⁺_{DPGi,Pr} neurons by microinjecting AAV-encoding Cre-dependent hM4Di-mCherry (AAV.8-*CamkIIa-DIO-hM4Di-mCherry*) into DPGi/Pr of *Nos1-Cre* mice (scale bar: 1 mm; yellow, hM4Di-mCherry; blue, DAPI).

(F) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral following administration of saline or CNO (3 mg/kg).

(G–J) Sleep architecture (G–I) and EEG power spectra (J) during the 3 h following saline or CNO (3 mg/kg) administration at ZT5. *n* = 12 mice for sleep architecture. *n* = 11 mice for EEG spectrum (1 mouse data was excluded due to contamination of large noises during wakefulness). **p* < 0.05; ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test). Data represent mean $\pm$ SEM.

(K) Fiber photometry recording of the population activity of *Nos1*⁺_{DPGi,Pr} neurons by microinjecting AAV-encoding Cre-dependent jRCaMP1a (AAV.9-*hSyn-flex-jRCaMP1a*) into the DPGi/Pr of *Nos1-Cre* mice (scale bar: 1 mm; green, jRCaMP1a; blue, DAPI).

(L) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, EMG integral, and photometry signal.

(M) Comparison of fluorescent signals among each vigilance state. Data represent mean $\pm$ SEM. *n* = 9 recording sessions obtained from 3 mice. **p* < 0.05, ††*p* < 0.01 (one-way ANOVA followed by Tukey's test, † indicates significant difference in one-way ANOVA, and * indicates significance in Tukey's test).

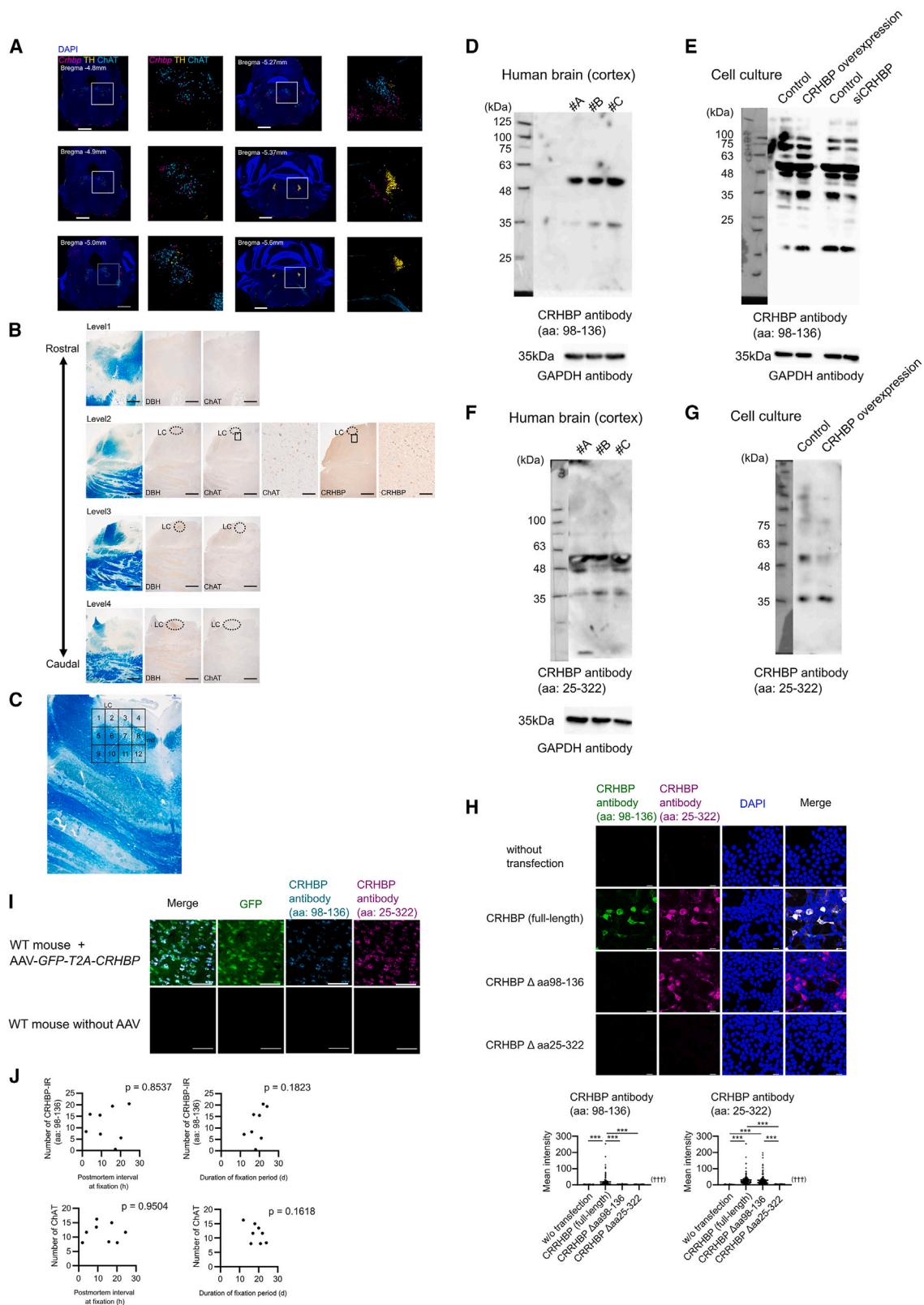
(N) Photometry signals aligned to vigilance state transitions. Data represent mean $\pm$ SEM. Wake to NREMS, *n* = 70 episodes; NREMS to wake, *n* = 48 episodes; NREMS to REMS, *n* = 84 episodes; REMS to wake, *n* = 20 episodes. All episodes were obtained from *n* = 9 recording sessions obtained from 3 mice. **p* < 0.05; ****p* < 0.001 (paired Student's *t* test).

(O) *Crh*⁺_{DPGi,Pr} neurons were genetically labeled by microinjecting AAV-encoding Cre-dependent hM3Dq-mCherry (AAV.2-*hSyn-DIO-hM3Dq-mCherry*) into DPGi/Pr of *Crh-Cre* mice, and co-expression of hM3Dq-mCherry (which labels *Crh*⁺_{DPGi,Pr} neurons) with *Vglut2* (ISH) or *Vgat* (ISH) was examined. Data represent mean $\pm$ SEM. *n* = 8 sections, obtained from 4 mice. (Scale bar: 0.2 mm; green, hM3Dq-mCherry; magenta, *Vgat* [ISH]; cyan, *Vglut2* [ISH]).

(P) The same mice were utilized to test the effects of chemogenetic activation of *Crh*⁺_{DPGi,Pr} neurons. Expression of hM3Dq-mCherry was detected in DPGi/Pr. (Scale bar: 0.5 mm; yellow, hM3Dq-mCherry; blue, DAPI).

(Q) Representative traces of EEG power spectrum, delta power, theta to delta power ratio, and EMG integral following administration of saline or CNO (1.5 mg/kg).

(R–U) Sleep architecture (R–T) and EEG power spectra (U) during the 3 h following saline or CNO (1.5 mg/kg) administration at ZT5 in mice expressing hM3Dq-mCherry in *Crh*⁺_{DPGi,Pr} neurons, *n* = 8 mice/group. **p* < 0.05; ***p* < 0.01; ****p* < 0.001 (paired Student's *t* test). Data represent mean $\pm$ SEM.



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Figure S6. Identification of the area homologous to the mouse SubLDT in humans and representative image of the area subjected to quantification of CRHBP-IR neurons and verification of CRHBP antibody, related to Figure 7

(A) (Mice) Distribution of *Crhbp*⁺ neurons, noradrenergic neurons (visualized by tyrosine hydroxylase [TH] immunostaining), and cholinergic neurons (visualized by ChAT immunostaining). At 5.0 and 5.27 mm, caudal to bregma, *Crhbp*⁺ neurons, noradrenergic neurons, and cholinergic neurons are intermingled within the SubLDT. Scale bar: 1 mm; magenta, *Crhbp* (ISH); yellow, TH (IHC); cyan, ChAT (IHC); blue, DAPI.

(B) (Human) Representative images of the upper portion of the left pons (5 mm interval between each level) derived from a control patient (#3) showing the distribution of CRHBP-IR neurons, noradrenergic neurons (visualized by DBH immunostaining), and cholinergic neurons (visualized by ChAT immunostaining). The locus coeruleus (LC), characterized by high density of DBH-IR noradrenergic neurons, is indicated as dotted circles. In level 2, cholinergic and CRHBP-IR neurons were observed in an area ventral to the most rostral part of LC. Klüver-Barrera staining (left), immunohistochemical staining for DBH, ChAT, or CRHBP. Scale bar: 1,000 μ m (low power field) and 100 μ m (high power field).

(C) To count CRHBP-IR neurons, 12 images were obtained from the area around SubLDT, and images showing CRHBP-IR neurons were used for further analysis. (D–G) Western blot image of human brain (cortex) (D and F) or cultured cell (E and G) homogenates using CRHBP antibodies used in this study or GAPDH antibody.

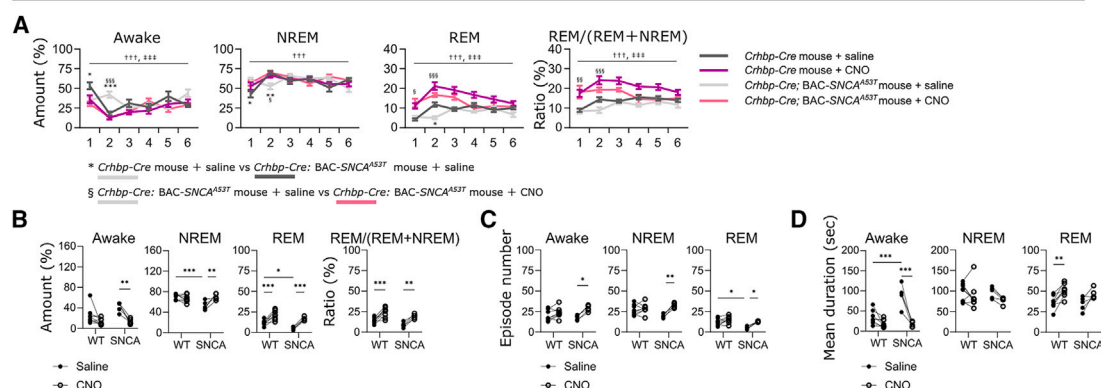
(H) Immunocytochemistry using CRHBP antibodies in HEK293T cells either without transfection or transfected with plasmids overexpressing full-length CRHBP, CRHBP Δ aa 98–136, or CRHBP Δ aa 25–322. Scale bar: 20 μ m. Green, CRHBP (aa 98–136) antibody; magenta, CRHBP (aa 25–322) antibody; blue, DAPI. Data represent mean $\pm$ SEM. $n = 164$ (without transfection), 165 (full-length CRHBP), 128 (CRHBP Δ aa 98–136), and 117 (CRHBP Δ aa 25–322) ROIs for CRHBP (aa 98–136) antibody. $n = 151$ (without transfection), 165 (full-length CRHBP), 116 (CRHBP Δ aa 98–136), and 141 (CRHBP Δ aa 25–322) ROIs for CRHBP (aa 25–322) antibody. ***,††† $p < 0.001$ (one-way ANOVA followed by Tukey's test, † indicates significant difference in one-way ANOVA, and * indicates significance in Tukey's test).

(I) IHC using CRHBP antibodies in brain sections of mice with or without microinjection into the thalamus of AAV that separately expresses GFP and full-length CRHBP (AAV.PHPeB-GFP-T2A-CRHBP). Scale bar: 0.1 mm; green, GFP (AAV infection marker); cyan, CRHBP (aa 98–136) antibody; magenta, CRHBP (aa 25–322) antibody.

(J) Correlation between the number of cells in immunohistochemistry and post-mortem interval at fixation (hours) or duration of fixation period (days), respectively. Pearson's r test.

See also Table S1 for detailed information about the clinical diagnosis and tissue preparation.

Crhbp-Cre; BAC-*SNCA*^{A53T} mouse; AAVrg-*DO-mCherry-DIO-flippase* (Gi) / AAV8-*fDIO-hM3Dq-YFP* (SubLDT)



A head-to-head comparison of the effect of ablation of *Crhbp*⁺_{SubLDT} and activation of *Crhbp*⁺_{SubLDT→Gi} neurons

Fig. S1

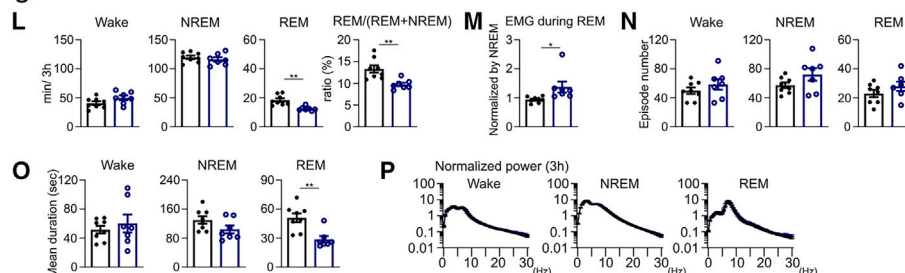
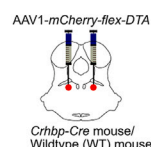


Fig. S2

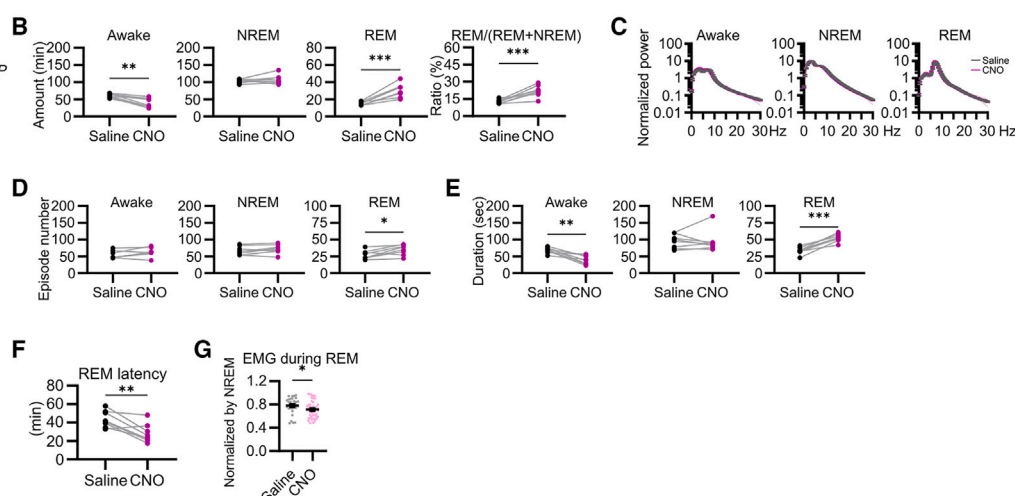
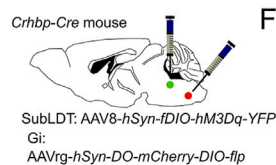


Figure S7. Analyses of the effect of chemogenetic activation of *Crhbp*⁺_{SubLDT→Gi} neurons in a mouse model of PD and a head-to-head comparison of Figures S1L–S1P and S2B–S2G, related to Figures 1, 3, and 7

Comparison of the effect of saline and CNO (1.5 mg/kg) administered at ZT5 on the amount of each vigilance state in *Crhbp-Cre* or *Crhbp-Cre*; BAC-*SNCA*^{A53T} mouse expressing hM3Dq in *Crhbp*⁺_{SubLDT→Gi} neurons. (A) Horizontal axes indicate the time after drug administration. (B–D) Sleep architecture during 1–2 h following administration. Data represent mean ± SEM. *n* = 8 mice in the *Crhbp-Cre* group and *n* = 5 mice in the *Crhbp-Cre*; *SNCA-Tg* group.

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*, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$ (two-way repeated-measures ANOVA followed by Bonferroni's test; ‡ and † indicate significant main effect of intervention and significant interaction between intervention and time, respectively, in two-way repeated-measures ANOVA, and *,§ indicates significance in Bonferroni's test).

Note that, to directly compare the effects of genetic ablation of *Crhbp*⁺_{SubLDT} neurons (Figures S1L–S1P) with the effects of chemogenetic activation of *Crhbp*⁺_{SubLDT→Gi} neurons (Figures S2B–S2G), the data from ZT5 to ZT8 were extracted.